

Objective Sublevels and Central-Path Hessian Conditioning

Abstract

The barrier Hessians in interior-point Newton systems can become ill-conditioned as the objective gap g tends to zero. We ask how much of this is forced by the feasible set and the objective alone. We consider a compact convex feasible set, a nonconstant linear objective, and the Hessian of a self-concordant barrier on the tangent space of the affine constraints, in a fixed Euclidean metric. Let $L(g)$ be the set of feasible points with gap at most g . For every barrier with finite parameter ν , the unit Dikin ellipsoid at the central-path point with gap g , translated to the origin, lies in the difference body $L(g) - L(g)$, which lies in the same ellipsoid scaled by $2(\nu + 2\sqrt{\nu})$. Hence Hessians of different barriers at equal gap agree in Loewner order up to parameter-dependent factors, and the condition number satisfies $\kappa(g) = \Theta((\text{diam } L(g)/g)^2)$ as $g \rightarrow 0$. These bounds extend to points with Newton decrement below one, with explicit constants that are uniform over bounded barrier parameters and decrements at most a fixed number below one. The j th eigenvalue of each Hessian lies between the inverse squares of two minimax (Courant–Fischer) radial widths of $L(g) - L(g)$, up to a parameter-dependent factor. For a trace-normalized version of a classical 3×3 semidefinite program of Sturm, whose optimality system has singularity degree two, every fixed barrier has condition number $\Theta(g^{-3/2})$, and we compute all four eigenvalues of the log-barrier Hessian with exact leading constants. Adding two linear constraints preserves the singularity degree but gives $\Theta(g^{-1})$. For linear programs, every fixed barrier Hessian has eigenvalues of orders $\Theta(1)$ and $\Theta(g^{-2})$ only; the eigenspace of the $\Theta(1)$ eigenvalues approaches the tangent space of the optimal face at rate $O(g)$, and $O(g^2)$ for the logarithmic barrier. An explicit barrier attains the rate $O(g)$; for it, discarding a right-hand-side component of vanishing relative size gives a relative Newton-direction error tending to one. We also bound the number of conjugate-gradient iterations, in exact arithmetic, for spectra in two clusters. The bounds concern this Hessian, not Schur-complement or augmented Newton matrices, and we give no running-time bounds for quantum linear-system solvers. Computations with exact rational certificates illustrate the results.

Keywords: self-concordant barriers; central path; Hessian conditioning; objective sublevels; semidefinite programming; spectral clustering.

1 Introduction

The Newton systems of interior-point methods are built from the Hessian of a self-concordant barrier, directly or through a Schur complement or an augmented system, and can become increasingly ill-conditioned as the requested objective accuracy improves. This is usually analyzed for the logarithmic barrier or for a particular primal–dual linear system. We ask how much of it is already forced by the feasible set and the objective. Can a different barrier, with a different central path, improve the growth rate of the condition number of the primal barrier Hessian? Which geometric features determine the individual eigenvalues? The answers matter when choosing a barrier and when interpreting solver costs that depend on the condition number.

We answer these questions for the primal barrier Hessian restricted to the tangent space of the affine constraints (the reduced Hessian), in a fixed Euclidean metric. Let P be a compact convex

feasible set with nonempty relative interior, and let the linear objective $\langle c, x \rangle$, to be minimized, be nonconstant on the affine hull of P . Let $L(g)$ be the set of feasible points whose objective value exceeds the optimal value by at most g (the primal objective gap), and let $D(g) = \text{diam } L(g)$. For a barrier F , let $\hat{H}_F(g)$ be the Hessian of F (not of μF) at the central point with gap g , that is, at the minimizer of $F(x) + \langle c, x \rangle / \mu$ whose gap is g . With κ the ratio of the largest to the smallest eigenvalue, our main conclusion is

$$\kappa(\hat{H}_F(g)) = \Theta\left(\frac{D(g)^2}{g^2}\right) \quad (g \downarrow 0) \quad (1)$$

for every fixed self-concordant barrier F with finite barrier parameter. The constants may depend on the fixed instance, the metric, and the barrier parameter, but not on g . Changing the barrier can change the constants and the central points, but not the rate, which is determined by the geometry of the objective sublevel sets.

The diameter law (1) follows from a stronger comparison. Let E_F be the unit Dikin ellipsoid of F at a point, taken in the tangent space, and let $K_g = L(g) - L(g)$ be the difference body of the sublevel set, which depends on neither the barrier nor the point. Suppose that, at a point with gap g , the Newton decrement ρ of $F + \langle c, x \rangle / \mu$, measured in the Hessian of F , is less than one for some $\mu > 0$. Then $E_F \subset K_g \subset 2CE_F$ at this point, where ν is the barrier parameter and

$$C = \frac{\nu + 2\sqrt{\nu} + \rho}{1 - \rho}, \quad 0 \leq \rho < 1.$$

Since K_g has diameter $2D(g)$, and the largest ball centered at the origin inside K_g has radius of order g , this containment gives (1).

1.1 Main results

1. *Comparison at equal objective accuracy.* Theorems 3.2, 3.3 and 4.1 prove the containment, the resulting Loewner-order comparison of the Hessians of different barriers at equal gap, the diameter law, and bounds on every eigenvalue: the j th eigenvalue lies between the inverse squares of two minimax radial widths of K_g , up to the factor $4C^2$. The estimates hold uniformly over barriers with bounded parameters and Newton decrements bounded uniformly below one (corollary 3.4). On an unbounded feasible set, the pointwise bounds hold at the points of any compact objective sublevel set (proposition 3.5). The finite barrier parameter is needed: section 10 gives a self-concordant function without one for which the law fails.
2. *Sharp geometric and spectral examples.* The law gives bounded conditioning at every unique optimum of a polytope, including degenerate vertices, and conditioning of order g^{-2} when the optimal face has positive dimension (proposition 5.2). For a fixed simplex and objectives c_ϑ whose optimum is a unique vertex for $\vartheta > 0$ and an edge at $\vartheta = 0$, the condition number is $\Theta(\min\{\vartheta^{-2}, g^{-2}\})$ uniformly in ϑ (proposition 5.4). For a trace-normalized version of a classical 3×3 semidefinite program, theorem 7.2 gives the asymptotics of all four eigenvalues of the reduced log-barrier Hessian, with exact constants. Adding two linear constraints gives a problem whose optimality system has the same singularity degree, two, but on which every fixed barrier has condition number $\Theta(g^{-1})$ instead of $\Theta(g^{-3/2})$ (proposition 7.1). Thus singularity degree alone does not determine the conditioning exponent.
3. *Newton right-hand sides, and what condition numbers do not show.* For a linear program with an optimal face of dimension f , every fixed barrier Hessian at the central point with

gap g has f eigenvalues of order one and the others of order g^{-2} . The orthogonal projector onto the eigenspace of the f small eigenvalues (the weak eigenspace) differs in norm from the projector onto the tangent space of the optimal face by $O(g)$, and by $O(g^2)$ for the logarithmic barrier (theorem 8.1). Hence, at an exact central point, the Newton right-hand side has a component of relative norm $O(g)$ in that eigenspace, and $O(g^2)$ for the logarithmic barrier. An explicit 20-self-concordant barrier on a rectangle attains the rate $O(g)$ along a sequence of gaps (proposition 8.2). For this barrier, discarding that right-hand-side component, although its relative size tends to zero, produces a solution whose relative Euclidean error tends to one. For spectra in two clusters, proposition 9.1 bounds the number of conjugate-gradient iterations in exact arithmetic.

1.2 Relation to prior work

The geometry of self-concordant barriers is classical (Nesterov and Nemirovskii, 1994; Renegar, 2001; Nesterov, 2018); we give short proofs of the facts we use (Dikin containment, semiboundedness, and asymmetric containment) to fix our constants. Nesterov and Nemirovskii (1994, Proposition 2.3.2(iii), equation (2.3.9)) compares the local Hessian norm with the symmetric chord radius at a fixed interior point, and so already compares the Hessians of different barriers at the *same point*. We compare Hessians at generally different points, by comparing each with the same difference body K_g , which does not depend on the point. Freund (2003) studies the geometry of primal–dual level sets. More directly, Xiong and Freund (2024, Section 5.1, Fact 5.2 and Remark 5.1) use central-path ellipsoids to relate the diameter and conic radius of primal–dual sublevel sets to Hessian eigenvalues, in order to analyze rescaling for PDHG; their work is a direct antecedent of our geometric method. We instead compare the primal Hessian on the tangent space with the difference body K_g at a prescribed primal gap, and obtain the diameter law and the comparison between barriers.

Renegar (1996) relates the conditioning of the linear equations of barrier methods to problem conditioning and to conjugate-gradient solution, and Peña (2001, 2002) develops a primal–dual condition-number approach and implicit-barrier methods. Peña (2002, Proposition 3.2) parameterizes the central path by the primal objective gap, and his Section 3.3 bounds the condition number of the Schur matrix $A\nabla^2 F(x)^{-1}A^\top$ by an inverse-square rate in the gap, under his assumptions. We therefore claim novelty neither for the gap parameterization nor for an inverse-square upper bound as such; for the reduced Hessian, we add the two-sided dependence on $D(g)$ and the comparison between barriers.

Spectral splitting of logarithmic-barrier Hessians and its numerical consequences were studied by Wright (1994, 1998); Forsgren et al. (2002) surveys interior methods, and Wright and Orban (2002) treat degenerate nonlinear programs. Limits of LP central paths and their analytic properties are classical (Adler and Monteiro, 1991; Güler, 1994; Halická, 1999); our limit of the log-barrier Hessian scaled by μ^2 at a unique LP optimum, including a degenerate vertex, is a corollary of this theory. Duistermaat (2001) analyzes the boundary asymptotics of the Hessian Riemannian structure under additional smoothness assumptions; his Remark 2.3 gives oscillatory self-concordant perturbations, and qualitative nonconvergence of central paths has earlier examples (Bolte and Pauwels, 2022). Our LP contributions are the eigenvalue orders for arbitrary finite-parameter barriers, the rates for the small-eigenvalue eigenspace, and, for the oscillatory example, the sharpness of the rate $O(g)$ and the calculation of the solution error.

For semidefinite programming, error bounds and facial reduction connect singularity degree with fractional powers (Sturm, 2000; Drusvyatskiy and Wolkowicz, 2017; Lourenço, 2021). We use a trace-normalized version of Sturm’s classical 3×3 example and do not claim a new source of error bounds with exponent $1/4$. Alizadeh et al. (1998) study a primal–dual Schur matrix, Wei

and Wolkowicz (2010) investigate instances where strict complementarity fails, and Sremac et al. (2021) relate singularity degree to primal and dual spectral rates along a regularization path whose constraints change with the path parameter. Our equality constraints, and hence the tangent space of our Hessian, are fixed; section 7 compares the paths and matrices in detail.

Finally, condition numbers alone do not determine solver cost. Cluster-sensitive conjugate-gradient analysis is classical (Axelsson, 1994); we give an explicit polynomial bound on the energy-norm error in exact arithmetic, in the form needed here. Quantum linear-system results also depend on access to the matrix, normalization, preparation of the right-hand side, and the required output (Gilyén et al., 2019; Orsucci and Dunjko, 2021), including in quantum interior-point methods such as the quantum speedups for LP of Apers and Gribling (2026). We do not derive an end-to-end quantum lower bound from the diameter law.

To the best of our knowledge, the equal-gap comparison between arbitrary finite-parameter barriers, with bounds that depend explicitly on the Newton decrement, and the resulting two-sided characterization by the diameter have not previously been stated in this form. Our other contributions are the explicit calculations listed above and their consequences for all barriers.

1.3 Organization

Sections 2 to 4 develop the general theory, sections 5 to 7 give classifications and exact examples, sections 8 and 9 treat Newton right-hand sides and linear solves, and section 10 explains the roles of the metric, the formulation, and the finite barrier parameter. Section 11 gives reproducible numerical illustrations with exact rational certificates of their input hypotheses; the claims they illustrate are proved earlier.

2 The objective gap and the reduced Hessian

In the main geometric results, P is a compact convex subset of a finite-dimensional Euclidean space, with affine hull $x^0 + \mathcal{V}$ and relative interior $P^\circ = \text{ri } P$. The tangent space $\mathcal{V}$ has dimension $d \geq 1$. We minimize a linear functional $\langle c, x \rangle$ that is nonconstant on P ; equivalently, its orthogonal projection $c_{\mathcal{V}}$ onto $\mathcal{V}$ is nonzero. Set

$$\text{OPT} = \min_{x \in P} \langle c, x \rangle, \quad \Phi = \{x \in P : \langle c, x \rangle = \text{OPT}\}, \quad (2)$$

$$g(x) = \langle c, x \rangle - \text{OPT}, \quad \Delta = \max_{x \in P} g(x) > 0. \quad (3)$$

For $0 \leq g \leq \Delta$, define

$$L(g) = \{y \in P : g(y) \leq g\}, \quad D(g) = \text{diam } L(g), \quad K_g = L(g) - L(g) \subset \mathcal{V}. \quad (4)$$

The difference body K_g consists of all chords of the sublevel set, translated to the origin. Unlike the diameter, it records the longest chord in every direction. For an interior point x , also put

$$\ell(x) = \max_{y \in L(g(x))} \|y - x\|, \quad \frac{D(g(x))}{2} \leq \ell(x) \leq D(g(x)). \quad (5)$$

The lower inequality follows from the triangle inequality applied to a pair of points at distance $D(g(x))$.

Definition 2.1 (Barrier and metric). A ν -self-concordant barrier for P , with $\nu > 0$, is a convex C^3 function $F : P^\circ \rightarrow \mathbb{R}$ that tends to $+\infty$ at every relative boundary point, has positive-definite tangent Hessian, and satisfies

$$|D^3F(x)[h, h, h]| \leq 2(D^2F(x)[h, h])^{3/2}, \quad (6)$$

$$|DF(x)[h]| \leq \sqrt{\nu}(D^2F(x)[h, h])^{1/2} \quad (h \in \mathcal{V}). \quad (7)$$

All gradients and Hessians below are taken along the affine space $x^0 + \mathcal{V}$, that is, in directions $h \in \mathcal{V}$. Write $H_F(x)$ for the self-adjoint operator on $\mathcal{V}$ representing $D^2F(x)$; we call it the reduced Hessian. Define

$$\|h\|_x = \sqrt{\langle H_F(x)h, h \rangle}, \quad \|v\|_x^* = \sqrt{\langle H_F(x)^{-1}v, v \rangle}, \quad E_F(x) = \{h \in \mathcal{V} : \|h\|_x \leq 1\}. \quad (8)$$

The condition number is $\kappa_F(x) = \lambda_{\max}(H_F(x))/\lambda_{\min}(H_F(x))$.

If W is an orthonormal basis matrix for $\mathcal{V}$, the matrix of this operator is $W^\top \nabla^2 F(x) W$; its eigenvalues do not depend on the choice of orthonormal basis. The Euclidean metric is fixed throughout. For symmetric matrix variables it is the Frobenius metric, and for several blocks it is the product Frobenius metric. We use the Hessian of F , not of μF . Multiplying by μ leaves the condition number unchanged but multiplies every eigenvalue by μ . A different representation of the feasible set, or a nonorthogonal change of coordinates, needs a separate analysis of the metric.

2.1 Two standard barrier facts, with proofs

The following facts are standard properties of self-concordant barriers (Nesterov and Nemirovskii, 1994; Renegar, 2001). We give short one-dimensional proofs to fix the normalization and the treatment of boundary points.

Lemma 2.2 (Dikin containment and semiboundedness). *For $x \in P^\circ$,*

$$x + \text{int}_{\mathcal{V}} E_F(x) \subset P^\circ, \quad x + E_F(x) \subset P, \quad (9)$$

and for every $y \in P$,

$$\langle \nabla F(x), y - x \rangle \leq \nu. \quad (10)$$

In particular, at every interior point, whether central or not,

$$\|c_{\mathcal{V}}\|_x^* \leq g(x). \quad (11)$$

Proof. Fix $h \in \mathcal{V}$ with $\|h\|_x = 1$, and write $\varphi(t) = F(x + th)$ for $0 \leq t < R$, where R is the first positive exit time from P° . Self-concordance gives

$$\left| \frac{d}{dt} \frac{1}{\sqrt{\varphi''(t)}} \right| \leq 1. \quad (12)$$

Thus $\varphi''(t) \leq (1 - t)^{-2}$ for $0 \leq t < \min\{R, 1\}$. If $R < 1$, integration twice bounds $\varphi(t)$ as $t \uparrow R$, contradicting boundary divergence. Hence $R \geq 1$. This proves the open containment, and closed containment follows by taking limits in the closed set P .

To prove (10), note that the case $y = x$ is immediate. Otherwise let $\psi(t) = F(x + t(y - x))$, $0 \leq t < 1$. These points are interior because P is convex and x is in the relative interior. If $\psi'(0) \leq 0$, the claim is immediate. If $\psi'(0) > 0$, convexity and (7) give

$$\psi'(t) > 0, \quad \psi''(t) \geq \frac{\psi'(t)^2}{\nu}, \quad \frac{1}{\psi'(s)} \leq \frac{1}{\psi'(0)} - \frac{s}{\nu} \quad (0 \leq s < 1).$$

The left side is positive. Letting $s \uparrow 1$ proves $\psi'(0) \leq \nu$, including boundary choices of y . Finally, minimizing the linear objective over the closed Dikin ellipsoid gives $\langle c, x \rangle - \|c_{\mathcal{V}}\|_x^* \geq \text{OPT}$. $\square$

2.2 Existence and objective-gap parameterization

For $\mu > 0$, the central point is

$$x_F(\mu) = \arg \min_{x \in P^\circ} \left(F(x) + \frac{\langle c, x \rangle}{\mu} \right), \quad \nabla F(x_F(\mu)) = -\frac{c_V}{\mu}. \quad (13)$$

We call $g_F(\mu) = g(x_F(\mu))$ the *primal objective gap* of the central point. Even for the logarithmic barrier, it need not equal the primal–dual gap $\nu\mu$.

Parameterizing central paths by the objective value is classical. In particular, the bounds in the following proposition agree with Peña (2002, Proposition 3.2). Our proof uses only the Dikin bound (11) and differentiation of the centrality condition.

Proposition 2.3 (The attained gap interval). *The analytic center $x_F^{\text{ac}} = \arg \min F$ and all points $x_F(\mu)$ exist and are unique. Put $a_F = g(x_F^{\text{ac}}) \in (0, \Delta)$. Then g_F is continuously differentiable, strictly increasing, and maps $(0, \infty)$ bijectively onto $(0, a_F)$. Moreover,*

$$g'_F(\mu) = \frac{\|c_V\|_{x_F(\mu)}^{*2}}{\mu^2} > 0, \quad (14)$$

$$\frac{\mu a_F}{\mu + a_F} \leq g_F(\mu) \leq \nu\mu. \quad (15)$$

Thus $g_F(\mu) = \Theta(\mu)$ as $\mu \downarrow 0$ for every fixed barrier.

Proof. A supporting affine function of F bounds it below on the compact set P . Since F diverges at the boundary, its bounded sublevel sets are compact subsets of P° . The same holds after adding any linear function. Existence follows, and the positive-definite Hessian gives strict convexity and uniqueness. The implicit function theorem applied to (13) gives $x'_F(\mu) = H_F(x_F(\mu))^{-1} c_V / \mu^2$, proving (14).

For any $x^* \in \Phi$, centrality and semiboundedness give

$$g_F(\mu) = \mu \langle \nabla F(x_F(\mu)), x^* - x_F(\mu) \rangle \leq \nu\mu.$$

Hence $g_F(\mu) \rightarrow 0$ as $\mu \downarrow 0$. For $\mu \rightarrow \infty$, optimality gives $F(x_F(\mu)) \leq F(x_F^{\text{ac}}) + \Delta/\mu$. Since a fixed sublevel set of F is compact and F has a unique minimizer, $x_F(\mu) \rightarrow x_F^{\text{ac}}$. An interior point cannot minimize or maximize a nonconstant linear functional, so $0 < a_F < \Delta$. Strict monotonicity proves the claimed range.

By (11) and (14), $(1/g_F)' \geq -1/\mu^2$. Integration from μ to M followed by $M \rightarrow \infty$ yields $1/g_F(\mu) \leq 1/a_F + 1/\mu$, which is the remaining inequality. $\square$

Let $\mu_F : (0, a_F) \rightarrow (0, \infty)$ be the inverse of g_F . We write the path parameterized by the gap, and its Hessian and condition number, as

$$\hat{x}_F(g) = x_F(\mu_F(g)), \quad \hat{H}_F(g) = H_F(\hat{x}_F(g)), \quad \hat{\kappa}_F(g) = \kappa_F(\hat{x}_F(g)). \quad (16)$$

Two fixed barriers can therefore be compared at equal gap on the common interval $(0, \min\{a_F, a_G\})$. Equal values of g generally correspond to different values of μ . This matters when the Hessians themselves, and not only their condition numbers, are compared.

3 The difference body determines the Hessian

We show that at an approximately central point of gap g the Dikin ellipsoid has the shape of the difference body K_g , up to explicit factors. First, the half of the Dikin ellipsoid on which the objective does not increase lies in the current sublevel set. Second, centrality places the whole sublevel set in a bounded dilation of that ellipsoid. Taking differences removes the center, so the Hessians of different barriers can be compared.

The containment facts are classical; [section 1](#) discusses prior work. What is new is the comparison with the difference body K_g , which applies at points of equal gap rather than at one point. It gives a Loewner comparison between different barriers at equal gap and, for every fixed barrier, two-sided bounds of order $(D(g)/g)^2$ on the condition number.

3.1 Containment with an explicit Newton decrement

For $x \in P^\circ$ and $\mu > 0$, define the Newton decrement (or centrality residual)

$$\rho_F(x; \mu) = \|\nabla F(x) + c_V/\mu\|_x^*. \quad (17)$$

This is the Newton decrement of $F + \langle c, \cdot \rangle / \mu$, whose Hessian is $H_F(x)$. We call x approximately central when this decrement is at most some $\rho < 1$; exact centers have $\rho = 0$. Define

$$C(\nu, \rho) = \frac{\nu + 2\sqrt{\nu} + \rho}{1 - \rho}, \quad C_\nu = C(\nu, 0) = \nu + 2\sqrt{\nu}. \quad (18)$$

Lemma 3.1 (One-sided containment). *Suppose $x \in P^\circ$, $y \in P$, and*

$$\langle \nabla F(x), y - x \rangle \geq -\rho \|y - x\|_x, \quad 0 \leq \rho < 1. \quad (19)$$

Then $\|y - x\|_x \leq C(\nu, \rho)$. Consequently, if $\rho_F(x; \mu) \leq \rho < 1$ for some $\mu > 0$ and $g = g(x)$, then

$$L(g) \subset x + C(\nu, \rho)E_F(x). \quad (20)$$

Proof. The assertion is immediate if $y = x$. Otherwise set $R = \|y - x\|_x$, $h = (y - x)/R$, and $\varphi(t) = F(x + th)$ for $0 \leq t < R$. The segment is interior up to its endpoint, even when y is on the boundary. Now $\varphi''(0) = 1$ and $\varphi'(0) \geq -\rho$. The other direction of [\(12\)](#) gives

$$\varphi''(t) \geq (1+t)^{-2}, \quad \varphi'(t) \geq \frac{t}{1+t} - \rho = \frac{(1-\rho)t - \rho}{1+t}. \quad (21)$$

Set $t_0 = (\sqrt{\nu} + \rho)/(1 - \rho)$. If $R \leq t_0$, the claimed bound follows because $t_0 < C(\nu, \rho)$. If $R > t_0$, semiboundedness at $x + t_0 h$, with endpoint y , gives

$$\varphi'(t_0)(R - t_0) \leq \nu, \quad \varphi'(t_0) \geq \frac{\sqrt{\nu}}{1 + t_0} > 0.$$

Therefore

$$R \leq t_0 + \sqrt{\nu}(1 + t_0) = \frac{\nu + 2\sqrt{\nu} + \rho}{1 - \rho}.$$

All estimates use derivatives only at interior points, so no derivative at y is required.

For the consequence, let $e = \nabla F(x) + c_V/\mu$. For $y \in L(g)$, $\langle c, y - x \rangle \leq 0$, and hence

$$\langle \nabla F(x), y - x \rangle = \langle e, y - x \rangle - \frac{\langle c, y - x \rangle}{\mu} \geq -\|e\|_x^* \|y - x\|_x.$$

Apply the first assertion. □

At $\rho = 0$ this is the usual asymmetric containment with constant $\nu + 2\sqrt{\nu}$; this constant is also used by [Xiong and Freund \(2024, Fact 5.2\)](#), citing [Nesterov \(2018, Theorem 5.3.8\)](#). The proof above extends it to every Newton decrement below one, without an exact center near x , and with the sublevel set taken at the gap $g(x)$ of x itself.

Theorem 3.2 (Difference-body and equal-gap comparisons). *Let $x \in P^\circ$ have gap g and satisfy $\rho_F(x; \mu) \leq \rho < 1$ for some $\mu > 0$. With $C = C(\nu_F, \rho)$ and $E = E_F(x)$,*

$$E \subset K_g \subset 2CE. \quad (22)$$

In particular, suppose barriers F, G have points x, y of the same gap g , with decrement bounds $\rho_F, \rho_G < 1$, respectively. Write $H_F = H_F(x)$, $H_G = H_G(y)$, and $C_F = C(\nu_F, \rho_F)$, $C_G = C(\nu_G, \rho_G)$. Then, on the fixed tangent space,

$$\frac{1}{4C_F^2} H_F \preceq H_G \preceq 4C_G^2 H_F. \quad (23)$$

The ordered eigenvalues obey the same inequalities. The condition numbers satisfy

$$\frac{\kappa_F(x)}{16C_F^2 C_G^2} \leq \kappa_G(y) \leq 16C_F^2 C_G^2 \kappa_F(x). \quad (24)$$

Proof. For $h \in E$, choose $\sigma \in \{-1, 1\}$ so that $\langle c, \sigma h \rangle \leq 0$. Closed Dikin containment gives $x + \sigma h \in L(g)$, while $x \in L(g)$. Thus $\sigma h \in K_g$. The difference body is symmetric, so $h \in K_g$. This inclusion requires no centrality. The reverse inclusion follows from (20): the difference of two vectors of local norm at most C has local norm at most $2C$.

At equal gap, $E_F(x) \subset K_g \subset 2C_G E_G(y)$. Applying this to an arbitrary nonzero vector normalized in the H_F norm proves $H_G \preceq 4C_G^2 H_F$. Interchanging F, G proves the other inequality. The eigenvalue inequalities follow from the variational principle, and their ratios give (24). $\square$

When the barrier changes, the central point with a given gap can move substantially. Nevertheless, the Dikin ellipsoids of the two barriers at their points of gap g have the same shape up to the factors in (23), because both approximate K_g . This controls every quadratic form and every ordered eigenvalue, not only the condition number. For exact centers it applies to every gap attained by both central paths ([proposition 2.3](#)).

3.2 Two-sided bounds on the condition number for every barrier

The difference-body comparison already determines the condition number through an aspect ratio of K_g , up to factors depending only on C . To express this in terms of the gap, fix a closed ball in the affine hull of P ,

$$z + r\mathbb{B}_{\mathcal{V}} \subset P, \quad r > 0, \quad (25)$$

where $\mathbb{B}_{\mathcal{V}}$ is the Euclidean unit ball in $\mathcal{V}$. Such a ball exists because P° is nonempty. Its radius and the objective range Δ do not depend on the barrier.

Theorem 3.3 (Aspect ratio, extreme eigenvalues, and the diameter law). *Under the hypotheses of [theorem 3.2](#), let*

$$r_g = \max\{s \geq 0 : s\mathbb{B}_{\mathcal{V}} \subset K_g\}.$$

Then $r_g > 0$ and

$$\left(\frac{D(g)}{2Cr_g}\right)^2 \leq \kappa_F(x) \leq \left(\frac{2CD(g)}{r_g}\right)^2. \quad (26)$$

More explicitly,

$$\ell(x)^{-2} \leq \lambda_{\min}(H_F(x)) \leq C^2 \ell(x)^{-2}, \quad (27)$$

$$\frac{\|c_V\|^2}{g^2} \leq \lambda_{\max}(H_F(x)) \leq \left(\frac{C\Delta}{rg}\right)^2, \quad (28)$$

$$\left(\frac{\ell(x)\|c_V\|}{Cg}\right)^2 \leq \kappa_F(x) \leq \left(\frac{C\Delta\ell(x)}{rg}\right)^2. \quad (29)$$

Consequently,

$$\frac{\|c_V\|^2}{4C^2} \left(\frac{D(g)}{g}\right)^2 \leq \kappa_F(x) \leq \frac{C^2\Delta^2}{r^2} \left(\frac{D(g)}{g}\right)^2. \quad (30)$$

The lower bounds for $\lambda_{\min}$ and $\lambda_{\max}$ hold at every interior point, without any centrality condition.

Proof. The Euclidean circumradius of K_g is $D(g)$. An ellipsoid with Hessian H has circumradius $\lambda_{\min}(H)^{-1/2}$ and inradius $\lambda_{\max}(H)^{-1/2}$. Apply both facts to (22) and divide the resulting inequalities to obtain (26).

For any Euclidean unit $h \in \mathcal{V}$, the vector $h/\sqrt{\langle H_F(x)h, h \rangle}$ is in $E_F(x)$. Choosing its objective-decreasing sign, as in the proof of theorem 3.2, shows that its Euclidean length is at most $\ell(x)$. Thus every unit-direction Rayleigh quotient is at least $\ell(x)^{-2}$. Conversely, choose $y \in L(g)$ with $\|y - x\| = \ell(x) > 0$. By (20), the Rayleigh quotient in direction $(y - x)/\ell(x)$ is at most $C^2/\ell(x)^2$. This proves (27).

The lower bound for $\lambda_{\max}$ follows from (11), since

$$g^2 \geq \langle H_F(x)^{-1}c_V, c_V \rangle \geq \frac{\|c_V\|^2}{\lambda_{\max}(H_F(x))}.$$

For the upper bound choose $x^* \in \Phi$. Because $0 < g < \Delta$, convexity and linearity give the homothetic inclusion

$$x^* + \frac{g}{\Delta}(P - x^*) \subset L(g), \quad \frac{2rg}{\Delta}\mathbb{B}_V \subset K_g. \quad (31)$$

The second inclusion follows by taking differences of the homothetic copy of the ball in (25). Together with $K_g \subset 2CE_F(x)$ it implies, for each Euclidean unit h , $\|(2rg/\Delta)h\|_x \leq 2C$. This is the upper bound in (28). Taking ratios proves (29), and (5) gives (30). $\square$

Geometrically, the inradius r_g is the minimum width of $L(g)$, that is, the smallest distance between two parallel supporting hyperplanes. Indeed, the support function of K_g in a unit direction h is

$$\max_{u \in L(g)} \langle h, u \rangle - \min_{v \in L(g)} \langle h, v \rangle.$$

A centered ball lies in a closed convex body exactly when its support function is at most that of the body in every direction. Taking the minimum over h proves the assertion. Since $x \in L(g)$ has gap g and $\Phi \subset L(g)$, the width in the direction of c_V is exactly $g/\|c_V\|$. Thus

$$\frac{2rg}{\Delta} \leq r_g \leq \frac{g}{\|c_V\|}. \quad (32)$$

The condition number is therefore the squared aspect ratio of K_g , up to factors depending only on C , and (32) turns this into the two-sided law (30).

Corollary 3.4 (Uniformity and accuracy rates). *For every fixed barrier on the fixed compact instance,*

$$\widehat{\kappa}_F(g) = \Theta((D(g)/g)^2) \quad (g \downarrow 0). \quad (33)$$

The same two-sided estimates hold uniformly for barriers that may depend on g , provided their parameters satisfy $\nu_F \leq \bar{\nu}$ and the points in question have Newton decrements at most $\bar{\rho} < 1$. For exact centers, all such barriers attain every gap in

$$0 < g < \frac{\Delta}{1 + C_{\bar{\nu}}}. \quad (34)$$

If $D(g) = \Theta(g^\theta)$ with $0 \leq \theta \leq 1$, every fixed barrier has $\widehat{\kappa}_F(g) = \Theta(g^{2\theta-2})$, and equivalently $\kappa_F(x_F(\mu)) = \Theta(\mu^{2\theta-2})$.

Proof. The first two assertions follow directly from (30) with $C \leq C(\bar{\nu}, \bar{\rho})$. At the analytic center, $\nabla F = 0$, so lemma 3.1 with $\rho = 0$ contains all of P in $x_F^{\text{ac}} + C_{\nu_F} E_F(x_F^{\text{ac}})$. If y maximizes the objective on P , then (11) gives

$$\Delta - a_F = \langle c, y - x_F^{\text{ac}} \rangle \leq C_{\nu_F} \|c_{\mathcal{V}}\|_{x_F^{\text{ac}}}^* \leq C_{\nu_F} a_F.$$

Thus $a_F \geq \Delta/(1 + C_{\nu_F})$, proving (34). The exponent follows by substitution and proposition 2.3. $\square$

The constants are explicit and separate three sources of dependence: the barrier parameter, the allowed Newton decrement, and the fixed geometry $(r, \Delta, \|c_{\mathcal{V}}\|)$. Uniformity over a family of problem instances also requires control of these geometric constants. We make no uniform claim when the barrier parameters are unbounded or the decrement bounds approach one. The law describes how the condition number grows as $g \downarrow 0$ on a fixed instance; it is not a runtime bound, and its constants may depend on the dimension.

3.3 Constants for canonical barriers, and unbounded feasible sets with compact sublevel sets

For the canonical LP and SDP barriers, a bound on the dual slacks sometimes improves the upper constant. The general law does not need this bound. Consider either $P = \{x : Ax = b, x \geq 0\}$ with $F = -\sum_i \log x_i$, or $P = \{X : AX = b, X_j \succeq 0 \ (j = 1, \dots, m)\}$ with $F = -\sum_j \log \det X_j$. Assume the respective strictly positive or positive-definite relative interior is nonempty, and use the Euclidean or product Frobenius metric. Let ν_{can} denote the number of scalar coordinates or the sum of matrix-block sizes. At a central point, the canonical dual slacks satisfy $s_i = \mu/x_i$ or $S_j = \mu X_j^{-1}$. Hence if, for all $0 < \mu \leq \bar{\mu}$, their largest coordinate or operator norm is at most M , then, with $g = g_{\text{can}}(\mu)$,

$$\lambda_{\max}(\widehat{H}_{\text{can}}(g)) \leq M^2 \mu^{-2}, \quad \widehat{\kappa}_{\text{can}}(g) \leq \left(\frac{M \nu_{\text{can}} \ell(\widehat{x}_{\text{can}}(g))}{g} \right)^2. \quad (35)$$

For the LP this follows by bounding $\sum_i h_i^2/x_i^2$; for the SDP, bound $\sum_j \|X_j^{-1/2} U_j X_j^{-1/2}\|_F^2$ by $\mu^{-2} M^2 \sum_j \|U_j\|_F^2$. Use $g \leq \nu_{\text{can}} \mu$ and (27) to obtain the condition-number bound.

Any strictly feasible point z or Z gives such a finite M . Let $\sigma > 0$ be its smallest coordinate or its smallest block eigenvalue. Centrality supplies a dual multiplier y_μ with $A^* y_\mu + s_\mu = c$ (using S_μ in the matrix case), where A^* is the adjoint of the equality-constraint map. Canonical complementarity gives $\langle b, y_\mu \rangle = \langle c, x_\mu \rangle - \nu_{\text{can}} \mu$. Consequently,

$$\langle z, s_\mu \rangle = \langle c, z \rangle - \langle b, y_\mu \rangle \leq g(z) + \nu_{\text{can}} \bar{\mu}, \quad M = \frac{g(z) + \nu_{\text{can}} \bar{\mu}}{\sigma} \text{ is a valid bound,}$$

with traces and block notation understood for the SDP. Since the slacks are positive, this bounds every coordinate or the operator norm. This constant requires neither nondegeneracy nor strict complementarity. By [corollary 3.4](#), the canonical barrier and every other fixed barrier have the same conditioning rate, and [\(23\)](#) compares their full spectra in both directions.

Boundedness of P is convenient for the existence of the central path and for defining Δ . The pointwise theorems need only bounded sublevel sets $L(g)$ for small g , as in standard-form instances with unbounded feasible regions.

Proposition 3.5 (Localization to a compact sublevel). *Let P instead be closed and convex, with nonempty relative interior, and let the nonconstant linear objective have a finite attained optimum. Suppose $L(\bar{g})$ is compact for some $\bar{g} > 0$. Let F satisfy [definition 2.1](#) on P° . There is a ball $z + r\mathbb{B}_{\mathcal{V}} \subset L(\bar{g})$ with $r > 0$. For every point $x \in P^\circ$ with $0 < g(x) \leq \bar{g}$ and $\rho_F(x; \mu) \leq \rho < 1$, all pointwise conclusions of [theorems 3.2](#) and [3.3](#) hold with Δ replaced by $\bar{g}$ in their upper bounds. The width bounds of [section 4](#) below hold as well. This assertion does not require a central path to exist on the unbounded set.*

Proof. Take $x^* \in \Phi$ and $z_0 \in P^\circ$. A point of the open segment from x^* to z_0 close enough to x^* gives $z \in P^\circ$ with $g(z) < \bar{g}$. A small relative ball about z lies in P and below the level $\bar{g}$. Dikin containment and semiboundedness are proved on line segments, so they remain valid. The set $L(g(x))$ is compact, so diameters and chord maxima are attained. The proofs above now apply, using

$$x^* + \frac{g}{\bar{g}}(L(\bar{g}) - x^*) \subset L(g), \quad \frac{2rg}{\bar{g}}\mathbb{B}_{\mathcal{V}} \subset K_g$$

in place of [\(31\)](#). All later width proofs use only these containments and bounded objective-decreasing rays in $L(g)$. $\square$

4 Directional widths determine the full spectrum

The diameter law uses only the largest and smallest scales of a sublevel set. The intermediate eigenvalues depend on how many independent directions have each scale. The radial widths of the difference body record this information without reference to a barrier or a center.

For $0 < g \leq \Delta$ and a Euclidean unit vector $h \in \mathcal{V}$, define the radial function of the difference body by

$$q_g(h) = \max\{t \geq 0 : th \in K_g\}. \quad (36)$$

The set K_g is compact, convex, symmetric, and contains a neighborhood of zero. Thus q_g is positive, finite, even, and continuous on the unit sphere. Continuity follows because q_g is the reciprocal of the Minkowski functional of K_g , a norm equivalent to the Euclidean norm. For $j = 1, \dots, d$, define the radial widths

$$w_j^-(g) = \inf_{\substack{U \subset \mathcal{V} \\ \text{codim } U = j-1}} \sup_{\substack{h \in U \\ \|h\|=1}} q_g(h), \quad (37)$$

$$w_j^+(g) = \sup_{\substack{S \subset \mathcal{V} \\ \text{dim } S = j}} \inf_{\substack{h \in S \\ \|h\|=1}} q_g(h). \quad (38)$$

Here and below U, S are linear subspaces. These two radial widths need not coincide for a general convex body.

Theorem 4.1 (Radial width bounds). *Under the hypotheses of [theorem 3.2](#), list the eigenvalues of $H_F(x)$ in nondecreasing order. Then for $j = 1, \dots, d$,*

$$\frac{1}{(w_j^+(g))^2} \leq \lambda_j(H_F(x)) \leq \frac{4C^2}{(w_j^-(g))^2}. \quad (39)$$

Moreover, $w_j^+(g) \leq w_j^-(g) \leq 2Cw_j^+(g)$, $w_1^+(g) = w_1^-(g) = D(g)$, and $w_d^+(g) = w_d^-(g) = r_g$.

Proof. The radial form of [\(22\)](#) is

$$\frac{1}{\sqrt{\langle H_F(x)h, h \rangle}} \leq q_g(h) \leq \frac{2C}{\sqrt{\langle H_F(x)h, h \rangle}}, \quad \|h\| = 1.$$

Equivalently, for the Rayleigh quotient $R(h) = \langle H_F(x)h, h \rangle$,

$$q_g(h)^{-2} \leq R(h) \leq 4C^2 q_g(h)^{-2}. \quad (40)$$

The two Courant–Fischer formulas in increasing eigenvalue order are

$$\lambda_j = \sup_{\text{codim } U=j-1} \inf_{h \in U, \|h\|=1} R(h), \quad \lambda_j = \inf_{\text{dim } S=j} \sup_{h \in S, \|h\|=1} R(h).$$

Apply the lower estimate in the second formula and the upper estimate in the first. Positivity and uniform boundedness away from zero justify taking reciprocal squared infima and suprema. Explicitly,

$$\begin{aligned} \lambda_j &\geq \inf_{\text{dim } S=j} \sup_{h \in S, \|h\|=1} q_g(h)^{-2} = \left(\sup_{\text{dim } S=j} \inf_{h \in S, \|h\|=1} q_g(h) \right)^{-2}, \\ \lambda_j &\leq 4C^2 \sup_{\text{codim } U=j-1} \inf_{h \in U, \|h\|=1} q_g(h)^{-2} = 4C^2 \left(\inf_{\text{codim } U=j-1} \sup_{h \in U, \|h\|=1} q_g(h) \right)^{-2}. \end{aligned}$$

These are [\(39\)](#).

Every j -dimensional S intersects every codimension- $(j-1)$ subspace U in a nonzero direction. A unit vector in that intersection shows $\inf_{h \in S} q_g(h) \leq \sup_{h \in U} q_g(h)$; taking the indicated extrema proves $w_j^+ \leq w_j^-$. Combining the two bounds in [\(39\)](#) gives $w_j^- \leq 2Cw_j^+$. For $j = 1$ both widths are the maximum radial extent, namely $D(g)$. For $j = d$ both are the minimum radial extent, namely the centered inradius r_g . $\square$

For an ellipsoid, both radial widths are its principal semiaxes in decreasing order, and the theorem reduces to the reciprocal-square relation between semiaxes and Hessian eigenvalues. For a general difference body, [\(39\)](#) is the corresponding variational statement. It can separate several spectral scales, which a single condition number cannot.

4.1 Exit distances along rays and a sharper constant at a given point

A second pair of widths, measured from x to the relative boundary along one ray per line direction, improves the factor $4C^2$ to C^2 . Fix $x \in P^\circ$. For a line $[h] \subset \mathcal{V}$ through the origin, choose its Euclidean unit representative $\hat{h}$ so that $\langle c, \hat{h} \rangle < 0$ whenever the objective is nonconstant on that

line. If the line is orthogonal to c_V , choose the sign for which the ray from x stays longer in P° , breaking ties arbitrarily. Define

$$\tau_x([h]) = \sup\{t \geq 0 : x + s\hat{h} \in P^\circ \text{ for } 0 \leq s < t\}. \quad (41)$$

The chosen ray stays in $L(g(x))$ until it leaves P° , so its exit point lies in $L(g(x))$, and $0 < \tau_x([h]) \leq \ell(x)$; this also holds in the unbounded setting of [proposition 3.5](#). The sign convention can make τ_x discontinuous at lines orthogonal to the objective; this is why the next definitions use infima and suprema. Put

$$v_j^-(x) = \inf_{\text{codim } U=j-1} \sup_{h \in U, \|h\|=1} \tau_x([h]), \quad (42)$$

$$v_j^+(x) = \sup_{\dim S=j} \inf_{h \in S, \|h\|=1} \tau_x([h]). \quad (43)$$

Corollary 4.2 (Exit-distance spectral bounds). *Under the hypotheses of [theorem 3.2](#),*

$$\frac{1}{(v_j^+(x))^2} \leq \lambda_j(H_F(x)) \leq \frac{C^2}{(v_j^-(x))^2}, \quad j = 1, \dots, d. \quad (44)$$

Also $v_j^+(x) \leq v_j^-(x) \leq Cv_j^+(x)$ and $v_1^+(x) = v_1^-(x) = \ell(x)$.

Proof. The exit point lies at local-norm distance at least one from x by open Dikin containment, and at most C by [\(20\)](#). Consequently, for every unit h ,

$$\tau_x([h])^{-2} \leq \langle H_F(x)h, h \rangle \leq C^2 \tau_x([h])^{-2}.$$

All exit times are bounded below by the radius of a fixed relative ball about x and above by $\ell(x)$. The minimax argument in the proof of [theorem 4.1](#) therefore applies, without continuity, and also proves $v_j^+ \leq v_j^-$. Combining the two bounds in [\(44\)](#) gives $v_j^- \leq Cv_j^+$. Finally, for any $y \in L(g(x))$ distinct from x , the ray along $y - x$ reaches at least as far as y . The selected sign agrees with that ray when the objective strictly decreases, and the longer-exit convention can only increase its length when the objective is constant. Thus $\sup_{[h]} \tau_x([h]) \geq \|y - x\|$. Maximizing over y and using the reverse inequality already noted proves $v_1^+ = v_1^- = \ell(x)$. $\square$

The radial widths depend only on the sublevel geometry, so they allow comparison across barriers. The exit-distance bounds depend on the iterate and give sharper constants there. Neither formulation assumes that the optimal face is nondegenerate or that the spectrum has only two clusters.

5 Conditioning regimes from objective geometry

The diameter law separates two questions: how quickly the objective sublevel sets shrink, and how their shape determines a barrier Hessian. Error bounds answer the first. Their connection to weak sharp minima and conic degeneracy is classical ([Burke and Ferris, 1993](#); [Sturm, 2000](#); [Loureço, 2021](#)). Here every two-sided estimate of $D(g)$ gives a two-sided conditioning rate for every fixed self-concordant barrier.

Corollary 5.1 (Error bounds and attained rates). *Suppose, for some $M > 0$, $\theta \in (0, 1]$, and all sufficiently small positive gaps,*

$$\text{dist}(y, \Phi) \leq Mg(y)^\theta. \quad (45)$$

Then

$$\max \left\{ \text{diam } \Phi, \frac{g}{\|c_V\|} \right\} \leq D(g) \leq \text{diam } \Phi + 2Mg^\theta. \quad (46)$$

If Φ is a singleton, every fixed barrier consequently has $\widehat{\kappa}_F(g) = O(g^{2\theta-2})$. More generally, whenever $D(g) = \Theta(g^\theta)$ for $\theta \in [0, 1]$,

$$\widehat{\kappa}_F(g) = \Theta(g^{2\theta-2}). \quad (47)$$

For a fixed barrier, the same rates hold with μ in place of g . The gap statements also hold for barriers with uniformly bounded parameters, at points with Newton decrements at most a fixed $\bar{\rho} < 1$.

Proof. For two points of $L(g)$, choose nearest points in the compact set Φ and apply the triangle inequality. Points already in Φ have zero distance, so this proves the upper bound for small g . The inclusion $\Phi \subset L(g)$ proves its first lower bound. For every $0 < g < \Delta$ there is a point $y \in P$ with gap exactly g , by taking a segment from an optimum to a maximizer. Then $g \leq \|c_V\| \|y - x^*\|$ for any $x^* \in \Phi$, which proves the second lower bound. Apply [theorem 3.3](#) and [proposition 2.3](#) to finish. $\square$

At a unique optimum, an error bound alone bounds the conditioning rate only from above. A matching lower bound requires sublevel chords whose lengths have the same exponent. This distinction matters for singularity-degree bounds, as the paired examples in [section 7](#) show.

Proposition 5.2 (Faces, vertices, and curved optima). *For every fixed barrier, the following statements hold.*

1. If $\dim \Phi > 0$, then $D(g) = \Theta(1)$ and $\widehat{\kappa}_F(g) = \Theta(g^{-2})$.
2. If P is a polytope and $\Phi = \{x^*\}$, then $D(g) = \Theta(g)$ and $\widehat{\kappa}_F(g) = \Theta(1)$. No nondegeneracy assumption on the vertex is needed.
3. If $\Phi = \{x^*\}$ and $D(g) = \Theta(\sqrt{g})$, then $\widehat{\kappa}_F(g) = \Theta(g^{-1})$.

Proof. A positive-dimensional optimal face has positive diameter, while $D(g) \leq \text{diam } P$. For the polytope assertion let $v_1, \dots, v_k$ be its vertices other than x^* and put $\delta = \min_i g(v_i) > 0$. Every point can be written $y = \alpha_0 x^* + \sum_i \alpha_i v_i$, with nonnegative coefficients summing to one. Thus

$$\|y - x^*\| \leq \text{diam}(P) \sum_i \alpha_i \leq \frac{\text{diam}(P)}{\delta} g(y).$$

This gives $D(g) = O(g)$; the lower bound in [\(46\)](#) gives the converse. All three conditioning claims now follow from [\(47\)](#). $\square$

These three cases are examples, not a classification of convex sets. The diameter law applies even when $D(g)$ does not behave like a power of g . For a fixed compact LP, however, the optimum is a vertex or a positive-dimensional face, so the asymptotic conditioning exponent is zero or two. An intermediate slope of $\log \widehat{\kappa}_F$ against $\log g$ over a finite range of gaps therefore does not establish a fractional asymptotic rate.

The polytope argument is the elementary finite-vertex form of weak sharpness. It also shows that primal degeneracy alone cannot make the condition number of the reduced Hessian unbounded as $g \downarrow 0$. The bound is per instance; its constant need not be small or uniform over the data. [Section 6](#) computes the bounded limit for the logarithmic barrier.

Example 5.3 (A curved singleton in a 2×2 spectrahedron). Minimize X_{22} over $X \succeq 0$, $\text{tr } X = 1$. Write

$$X = \begin{pmatrix} 1-u & v \\ v & u \end{pmatrix}, \quad 0 \leq u \leq 1, \quad v^2 \leq u(1-u).$$

The unique optimum is $\text{Diag}(1, 0)$. If $u \leq g$, the Frobenius distance to this optimum is $O(\sqrt{g})$. The two feasible points with $u = g$ and $v = \pm\sqrt{g(1-g)}$ are separated by $2\sqrt{2g(1-g)}$, so $D(g) = \Theta(\sqrt{g})$. Hence every fixed barrier has condition number $\Theta(g^{-1})$.

For $F = -\log \det X$, symmetry forces $v = 0$ on the central path. Centrality in u gives

$$\mu = \frac{g(1-g)}{1-2g}, \quad 0 < g < \frac{1}{2}.$$

The coordinate directions $\partial X / \partial u$ and $\partial X / \partial v$ are orthogonal and both have squared Frobenius norm two. The two eigenvalues of the tangent Hessian are therefore

$$\frac{1}{g(1-g)}, \quad \frac{1}{2}(g^{-2} + (1-g)^{-2}).$$

They are in increasing order for small g , and $\hat{\kappa}_{\log}(g) \sim (2g)^{-1}$.

5.1 A uniform plateau near a nearly optimal edge

Consider the fixed simplex with a varying objective,

$$P = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1, x \geq 0\}, \quad c_\vartheta = (0, \vartheta, 1), \quad 0 < \vartheta \leq 1. \quad (48)$$

For each positive ϑ the unique optimum is e_1 ; at $\vartheta = 0$ the optimal set becomes an edge. The family separates the asymptotic rate, which is bounded, from a large constant caused by the nearly optimal edge. As g decreases, the condition number grows like g^{-2} until $g \approx \vartheta$ and then stays of order ϑ^{-2} , the plateau ([proposition 5.4](#)).

Proposition 5.4 (The two gap scales of the simplex). *Uniformly for $0 < \vartheta \leq 1$ and $0 < g \leq 1$,*

$$D_\vartheta(g) = \Theta(\min\{g/\vartheta, 1\}). \quad (49)$$

For any family of barriers with parameters at most $\bar{\nu}$, and for all g below a threshold independent of ϑ ,

$$\hat{\kappa}_{F,\vartheta}(g) = \Theta(\min\{\vartheta^{-2}, g^{-2}\}), \quad (50)$$

with constants depending only on $\bar{\nu}$. The same estimate holds at approximately central points, where they exist, with a common Newton-decrement bound $\bar{\rho} < 1$; the constants then also depend on $\bar{\rho}$.

Proof. For $x \in L_\vartheta(g)$, $\vartheta(x_2 + x_3) \leq \vartheta x_2 + x_3 \leq g$, while $x_2 + x_3 \leq 1$. Hence

$$\|x - e_1\| \leq \sqrt{2} \min\{g/\vartheta, 1\}.$$

The edge point $(1-t, t, 0)$, $t = \min\{g/\vartheta, 1\}$, belongs to $L_\vartheta(g)$ and has distance $\sqrt{2}t$ from e_1 . Taking diameters proves (49) with absolute constants. The set P and its interior-ball radius are fixed, its objective range is $\Delta = 1$, and

$$\|(c_\vartheta)_\nu\|^2 = \frac{2}{3}(\vartheta^2 - \vartheta + 1) \in [1/2, 2/3].$$

The diameter law is therefore uniform in ϑ . The common attained gap interval follows from [corollary 3.4](#). $\square$

For the logarithmic barrier the limiting plateau height is explicit:

$$\lim_{\mu \downarrow 0} \kappa_{\log, \vartheta}(x_{\log, \vartheta}(\mu)) = \frac{1 + \vartheta^2 + \sqrt{1 - \vartheta^2 + \vartheta^4}}{1 + \vartheta^2 - \sqrt{1 - \vartheta^2 + \vartheta^4}}, \quad (51)$$

and multiplying this limit by ϑ^2 and then letting $\vartheta \downarrow 0$ gives $4/3$. To prove this, write the equality multiplier as $\alpha_\mu = \mu/x_1 > 0$. Centrality gives

$$x_2 = \frac{\mu}{\vartheta + \alpha_\mu}, \quad x_3 = \frac{\mu}{1 + \alpha_\mu}, \quad \alpha_\mu \rightarrow 0.$$

The last limit follows from $x(\mu) \rightarrow e_1$, itself implied by $g(\mu) \rightarrow 0$. On the tangent vector $h = (-u - v, u, v)$ the Rayleigh quotient of $\mu^2 H_{\log}$ converges to

$$\frac{\vartheta^2 u^2 + v^2}{2(u^2 + uv + v^2)}.$$

Its eigenvalues in the fixed Euclidean metric are $(1 + \vartheta^2 \pm \sqrt{1 - \vartheta^2 + \vartheta^4})/3$, which proves (51). Their product is $\vartheta^2/3$, and the larger tends to $2/3$ as $\vartheta \downarrow 0$, proving the stated $4/3$ limit. This is an iterated limit: μ first tends to zero with ϑ fixed. The jointly uniform statement is (50), which describes the transition near $g \asymp \vartheta$ without interchanging limits.

6 The limiting Hessian at a unique LP optimum

At a unique optimum of a polytope, the diameter law gives bounded conditioning for every fixed barrier. For the logarithmic barrier, μ^2 times the Hessian converges to a positive-definite matrix, even at a degenerate vertex. We present this as a corollary of classical LP central-path endpoint theory, not as a new convergence theorem. Adler and Monteiro (1991, Theorem 3.3) identify the dual limit as the analytic center of the dual optimal face; their Theorem 5.1 and the discussion after it explicitly cover a unique optimum, where the columns of A on its support are linearly independent. Weighted path limits and analytic extension were further developed by Güler (1994) and Halická (1999). We state the consequence explicitly for the Hessian on $\ker A$.

Let

$$P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}, \quad A \in \mathbb{R}^{m \times n}, \quad \text{rank } A = m < n, \quad (52)$$

be compact and contain a strictly positive point. Suppose $c^\top x$ has a unique optimum x^* . Use $F_{\log}(x) = -\sum_i \log x_i$, and let the columns of $W \in \mathbb{R}^{n \times (n-m)}$ be an orthonormal basis of $\ker A$. The centrality equations provide positive slacks $s(\mu)$ and multipliers $y(\mu)$ with

$$Ax(\mu) = b, \quad A^\top y(\mu) + s(\mu) = c, \quad x_i(\mu)s_i(\mu) = \mu. \quad (53)$$

In particular, the dual problem is strictly feasible. Define $B = \{i : x_i^* > 0\}$ and $N = \{1, \dots, n\} \setminus B$. Classical endpoint theory gives

$$s(\mu) \rightarrow s^a, \quad s_B^a = 0, \quad s_N^a > 0. \quad (54)$$

Here s^a is the analytic center of the dual optimal slack face. The strict inequality on N is strict complementarity of the central-path limits; it does not assert primal nondegeneracy.

Proposition 6.1 (A classical endpoint corollary). *For the data above,*

$$\mu^2 H_{\log}(x(\mu)) \longrightarrow L := W^\top \text{Diag}((s^a)^2)W \succ 0, \quad (55)$$

where matrices represent operators in the basis W . Therefore

$$\mu^2 \lambda_j(H_{\log}(x(\mu))) \longrightarrow \lambda_j(L) \quad (1 \leq j \leq n-m), \quad \kappa_{\log}(x(\mu)) \longrightarrow \kappa(L) < \infty. \quad (56)$$

The conclusion includes the degenerate-vertex case $|B| < m$.

Proof. Complementarity yields the exact identity

$$\mu^2 W^\top \text{Diag}(x(\mu)^{-2})W = W^\top \text{Diag}(s(\mu)^2)W.$$

The right side converges in operator norm by (54). To prove that its limit is positive definite, first observe that A_B has full column rank. Otherwise a nonzero $v \in \ker A_B$ gives feasible points $(x_B^* \pm tv, 0_N)$ for sufficiently small $t > 0$. Since $c_B = A_B^\top y^a$ by complementary slackness, both points have the optimal objective value, contradicting uniqueness.

If $z \in \ker A$ has $z^\top \text{Diag}((s^a)^2)z = 0$, positivity of s_N^a gives $z_N = 0$. Then $A_B z_B = 0$ and full column rank imply $z = 0$. Hence the quadratic form is positive definite on $\ker A$. Continuity of ordered eigenvalues and of the condition number on positive-definite matrices proves (56). $\square$

6.1 An active-set formula and quantitative constants

Write a_i for column i of A . The limiting slack can be computed from the active set without following the central path. It is $s^a = c - A^\top y^a$, where

$$\begin{aligned} y^a &= \arg \max_y \sum_{i \in N} \log(c_i - a_i^\top y) \\ \text{subject to } & A_B^\top y = c_B, \quad c_N - A_N^\top y > 0. \end{aligned} \quad (57)$$

Every y feasible for (57) is dual optimal, because $b = A_B x_B^*$. Existence and uniqueness of this maximizer also follow directly once strict complementarity is known. If $z > 0$ is a fixed primal feasible point and s is dual optimal, then $z^\top s = c^\top z - \text{OPT}$ bounds every coordinate of s . Full row rank makes $y \mapsto A^\top y$ injective, so the closed dual optimal face is bounded. The logarithmic objective tends to $-\infty$ as an N -slack tends to zero. On the tangent space $\ker A_B^\top$, its negative Hessian is

$$A_N \text{Diag}(s_N^{-2}) A_N^\top.$$

It is positive definite on that tangent space: if both $A_N^\top h$ and $A_B^\top h$ vanish, full row rank gives $h = 0$. Thus the maximizer is unique; if the tangent space is zero-dimensional, it is the single feasible multiplier.

Let R_N restrict a vector to its N -coordinates, and define

$$\sigma_N = \sigma_{\min}(R_N W) > 0, \quad \eta_N = \|R_N W\|_2 \leq 1, \quad s_{\min} = \min_{i \in N} s_i^a, \quad s_{\max} = \max_{i \in N} s_i^a.$$

The uniqueness argument in the proof of [proposition 6.1](#) shows that $R_N W$ is injective. The bounds

$$s_{\min}^2 (R_N W)^\top (R_N W) \preceq L \preceq s_{\max}^2 (R_N W)^\top (R_N W) \quad (58)$$

give

$$s_{\min}^2 \sigma_N^2 \leq \lambda_{\min}(L) \leq s_{\max}^2 \sigma_N^2, \quad (59)$$

$$s_{\min}^2 \eta_N^2 \leq \lambda_{\max}(L) \leq s_{\max}^2 \eta_N^2, \quad (60)$$

$$\left(\frac{s_{\min}}{s_{\max}}\right)^2 \left(\frac{\eta_N}{\sigma_N}\right)^2 \leq \kappa(L) \leq \left(\frac{s_{\max}}{s_{\min}}\right)^2 \left(\frac{\eta_N}{\sigma_N}\right)^2. \quad (61)$$

These bounds separate the spread of the limiting slacks s_N^a from the angle between $\ker A$ and the coordinate subspace supported on B . Uniqueness of the optimum controls neither. The simplex in [proposition 5.4](#) already shows that the limiting condition number is unbounded over compact instances of fixed dimension.

Example 6.2 (A compact degenerate vertex). Take

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3/2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 3/2 \end{pmatrix}, \quad c = (1, 1, 0, 1)^\top. \quad (62)$$

The first equality constraint places P in the simplex, so P is compact. The point $(2, 3, 3, 3)^\top/11$ is strictly feasible, and $y = (-1, 0)^\top$ is strictly dual feasible. The unique optimum is e_3 , so $|B| = 1 < m = 2$. The dual optima have the form $y = (-3t/2, t)^\top$, with

$$(s_1, s_2, s_4) = (1 + 3t/2, 1 + t/2, 1 - 3t/2), \quad -2/3 < t < 2/3$$

for the relative interior of the dual optimal face. Setting the derivative of the sum of the three logarithms to zero gives $27t^2 + 36t - 4 = 0$. Only one root lies in the displayed interval:

$$t_a = \frac{-6 + 4\sqrt{3}}{9}. \quad (63)$$

Strict concavity makes it the dual analytic center. For an exact calculation without choosing an orthonormal nullspace basis, write

$$z = Tu, \quad T = \begin{pmatrix} -1/3 & 1 \\ 1 & 0 \\ -2/3 & -2 \\ 0 & 1 \end{pmatrix}, \quad T^\top T = \begin{pmatrix} 14/9 & 1 \\ 1 & 6 \end{pmatrix}.$$

The columns of T span $\ker A$. The eigenvalues of L are the generalized eigenvalues of

$$T^\top \text{Diag}((s^a)^2) T = \begin{pmatrix} (s_1^a)^2/9 + (s_2^a)^2 & -(s_1^a)^2/3 \\ -(s_1^a)^2/3 & (s_1^a)^2 + (s_4^a)^2 \end{pmatrix}$$

against $T^\top T$. Numerically these are approximately 0.24888160 and 1.14289062, giving $\kappa(L) \approx 4.59210571$. The exact matrix formulas establish the finite limit; the decimals only indicate its size.

The same endpoint proof applies to a strictly primal–dual feasible LP with an unbounded feasible region, whenever the central path has the limits stated above. For example, for $M \geq 1$,

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad c = (0, 1, 1, M)^\top,$$

the unique optimum e_1 is degenerate and $x(\mu) = (1, \mu, \mu, \mu/M)^\top$, $s(\mu) = (\mu, 1, 1, M)^\top$ solve the centrality equations exactly. In the orthonormal tangent basis $(0, 1, 1, 0)^\top/\sqrt{2}$, $(0, 0, 0, 1)^\top$, one has $\mu^2 H_{\log} = \text{Diag}(1, M^2)$. Every sublevel set with positive gap is compact, so [proposition 3.5](#) also applies to central points with small gap. Thus, at a degenerate vertex, the finite limit can have an arbitrarily large condition number $\kappa(L) = M^2$.

7 Fractional spectra and the limits of singularity degree

For a polytope, the sublevel diameter $D(g)$ is either of order one or of order g (proposition 5.2). Semidefinite sublevels can shrink at other rates. We analyze a trace-normalized version of the 3×3 instance in Sturm (2000, Example 2). Its nested constraints and quarter-power error bound are classical; see also Drusvyatskiy and Wolkowicz (2017, Theorem 4.5.1, Example 4.5.2, and Section 4.7). We compute its entire reduced log-barrier spectrum, extend the eigenvalue orders to every fixed barrier, and compare it with a restricted problem that has the same singularity degree but a different conditioning exponent.

7.1 Two spectrahedra and their sharp diameters

Consider

$$\min X_{22} \quad \text{subject to} \quad X \in \mathbb{S}_+^3, \quad \text{tr } X = 1, \quad X_{33} = X_{12}, \quad (64)$$

and the restricted problem obtained by adding

$$X_{13} = X_{23} = 0. \quad (65)$$

Here $\mathbb{S}_+^3$ is the cone of real symmetric positive semidefinite matrices. Write their feasible sets as P and P_0 , and their sublevel diameters as D and D_0 , respectively, each in its inherited Frobenius metric.

Proposition 7.1 (Equal singularity degree, different conditioning). *Both problems have compact feasible sets with positive-definite feasible points and unique optimum $X^* = e_1 e_1^\top$. The optimality systems of both problems have singularity degree two. Nevertheless,*

$$D(g) = \Theta(g^{1/4}), \quad D_0(g) = \Theta(g^{1/2}). \quad (66)$$

Thus every fixed self-concordant barrier on the respective set satisfies

$$\hat{\kappa}_F(g) = \Theta(g^{-3/2}) \quad \text{on } P, \quad \hat{\kappa}_{F_0}(g) = \Theta(g^{-1}) \quad \text{on } P_0. \quad (67)$$

For the displayed conic formulations, neither problem has a strictly complementary optimal primal-dual pair.

We use the standard definition of the singularity degree of a feasible conic *system*: the minimum number of facial-reduction steps needed to reach a face whose relative interior contains a point that satisfies the affine constraints (Drusvyatskiy and Wolkowicz, 2017; Waki and Muramatsu, 2013). At each step, an equality combination that is nonnegative on the current cone and zero on the affine system exposes a smaller face. In proposition 7.1, the system includes the optimality equation $X_{22} = 0$. Without this equation, the feasible systems have singularity degree zero, because they contain positive-definite matrices. This distinction matters when error bounds are applied to objective accuracy.

Proof. Trace normalization gives compactness. The matrix

$$\begin{pmatrix} 2/5 & 1/5 & 0 \\ 1/5 & 2/5 & 0 \\ 0 & 0 & 1/5 \end{pmatrix}$$

is positive definite and feasible for both problems. Write a point of P as

$$X = \begin{pmatrix} a & b & t \\ b & u & e \\ t & e & b \end{pmatrix}, \quad a + u + b = 1. \quad (68)$$

If $u = 0$, the second row and column of a PSD matrix vanish, so $b = e = 0$. The third diagonal is then zero, so $t = 0$. Thus the unique optimum in both sets is X^* , with value zero.

For $X \in L(g)$, principal minors give

$$0 \leq b \leq \sqrt{g}, \quad t^2 \leq ab \leq \sqrt{g}, \quad e^2 \leq ub \leq g^{3/2}, \quad |a - 1| = u + b \leq g + \sqrt{g}. \quad (69)$$

Together with $0 \leq u \leq g$, these inequalities prove $\|X - X^*\|_F = O(g^{1/4})$. If $t = e = 0$, they instead give $\|X - X^*\|_F = O(g^{1/2})$. These prove both diameter upper bounds.

For $0 < g \leq 1/16$, set

$$u = g, \quad b = \frac{\sqrt{g}}{2}, \quad a = 1 - g - \frac{\sqrt{g}}{2}, \quad e = 0, \quad t = \pm \frac{g^{1/4}}{2\sqrt{2}}. \quad (70)$$

Here $a \geq 13/16$ and $ag - b^2 = g(a - 1/4) > 0$, while

$$\det X = agb - b^3 - gt^2 = g^{3/2}(a/2 - 1/4) > 0.$$

The leading principal minors are positive, so both matrices are positive definite. Their Frobenius distance is $g^{1/4}$, proving the first lower bound. For P_0 , take the same a, b, u and $t = e = 0$. The resulting matrix is positive definite and its distance from X^* is at least $\sqrt{2}b = \sqrt{g/2}$, proving the second.

For the facial-reduction claim, let $E_{ii} = e_i e_i^\top$ and $J_{ij} = (e_i e_j^\top + e_j e_i^\top)/2$, so that $\langle J_{ij}, X \rangle = X_{ij}$. The first equality $X_{22} = 0$ exposes the face supported on $\text{span}\{e_1, e_3\}$. On this face $X_{12} = 0$, so $X_{33} - X_{12} = 0$ exposes the ray generated by E_{11} . Its relative interior contains X^* , proving that two steps suffice in both systems.

To show that one step does not suffice, note that an exposing combination for either optimality system has the form

$$Z = \alpha I + \beta(E_{33} - J_{12}) + \gamma E_{22} + \delta J_{13} + \eta J_{23} \succeq 0, \quad (71)$$

where $\delta = \eta = 0$ for P . Its value on the affine system must be zero. The trace equation is the only equation with a nonzero right side, so $\alpha = 0$; equivalently, $Z_{11} = 0$. A PSD matrix with zero diagonal entry has a zero corresponding row. The first row therefore forces $\beta = \delta = 0$; then $Z_{33} = 0$ forces $\eta = 0$. Consequently every nonzero exposing combination in the first step is a positive multiple of E_{22} . It cannot expose the final ray. This proves minimality of the two-step chain.

The same calculation establishes the complementarity assertion. A dual slack has the form (71) with $\gamma = 1$, since the objective is X_{22} ; optimality makes $\alpha = 0$. The only optimal slack is therefore E_{22} . It has rank one, as does X^* , so their ranks sum to two rather than three. Dual strict feasibility nonetheless holds: choosing a negative multiplier for the trace equation and all other multipliers zero makes the slack $E_{22} + \tau I \succ 0$ for $\tau > 0$. Finally, [theorem 3.3](#) proves (67) for every fixed barrier. $\square$

For an SDP with a compact feasible set and a unique optimum, if the optimality system has singularity degree s , the classical error bound gives $D(g) = O(g^{2-s})$ ([Sturm, 2000](#); [Drusvyatskiy and Wolkowicz, 2017](#)). Indeed, a feasible point has zero cone residual and violates only the added optimality equation, by g . A right inverse of the equality map on its range shows that the distance

from the point to the optimal affine space is at most a fixed multiple of g . The error bound then controls its distance to the optimum. If this bound is attained, that is, $D(g) = \Theta(g^{2^{-s}})$, the diameter law ([theorem 3.3](#)) gives

$$\widehat{\kappa}_F(g) = \Theta(g^{2^{1-s}-2}). \quad (72)$$

The two degree-two systems above show that attainment must be verified: the first has the quarter-power diameter, whereas the second has the faster square-root decay. Thus singularity degree alone does not determine the conditioning exponent, even for these compact 3×3 problems.

7.2 The exact log-barrier center and four spectral scales

[Theorem 3.3](#) gives the condition-number exponent on P . In this example we can also compute the order and the leading constant of every eigenvalue of the reduced log-barrier Hessian.

Theorem 7.2 (Four reduced-Hessian scales). *For $F(X) = -\log \det X$ on (64), the central point at gap g has the form*

$$\widehat{X}_{\log}(g) = \begin{pmatrix} a & b & 0 \\ b & g & 0 \\ 0 & 0 & b \end{pmatrix}, \quad b = \frac{-g + \sqrt{3g - 2g^2}}{3}, \quad a = 1 - g - b, \quad (73)$$

and the corresponding central-path parameter satisfies

$$q = ag - b^2, \quad \mu = \frac{q}{1 - b - 2g}. \quad (74)$$

These formulas apply for all sufficiently small positive g . The four eigenvalues of the reduced Hessian in the Frobenius metric, in increasing order, satisfy, as $\mu \downarrow 0$,

$$\lambda_1 \sim \sqrt{2}\mu^{-1/2}, \quad \lambda_2 \sim \mu^{-1}, \quad \lambda_3 \sim \sqrt{2}\mu^{-3/2}, \quad \lambda_4 \sim \frac{4}{7}\mu^{-2}. \quad (75)$$

In particular,

$$\kappa_{\log}(X_{\log}(\mu)) \sim \frac{2\sqrt{2}}{7}\mu^{-3/2} \sim \frac{3\sqrt{3}}{7}g^{-3/2}. \quad (76)$$

On P_0 , the log-barrier center is the same matrix. The two eigenvalues of its reduced Hessian are asymptotic to μ^{-1} and $(4/7)\mu^{-2}$, respectively, so its condition number is asymptotic to $(4/7)\mu^{-1}$, or $(6/7)g^{-1}$.

Proof. Conjugation by $\text{Diag}(1, 1, -1)$ preserves the feasible set, the objective, and F . Uniqueness of the central point therefore forces $t = e = 0$ in (68). At a fixed gap $u = g$, the remaining free coordinate is b , and

$$F = -\log b - \log q, \quad q = g(1 - g - b) - b^2.$$

Stationarity in b gives $q + bq_b = 0$, or

$$3b^2 + 2gb + g^2 - g = 0.$$

The positive root is the value in (73). Stationarity of $g/\mu + F$ in g gives $1/\mu = q_g/q$, which is (74). For small $g > 0$, $b > 0$, $a > 0$, $q > 0$, and $q_g > 0$. Thus these matrices are interior stationary points for a positive μ , hence are the unique central points by strict convexity. Conversely, uniqueness and the fixed-gap stationarity determine every small-gap center. The same argument holds on P_0 , where $t = e = 0$ are imposed as constraints.

To parametrize the whole central path for small gaps (not only a sequence of central points), put $\tau = g/b$ and $d = \tau^2 + 2\tau + 3$. The stationarity equations become

$$a = \frac{\tau + 3}{d}, \quad b = \frac{\tau}{d}, \quad g = \frac{\tau^2}{d}, \quad q = \frac{\tau^2(\tau + 2)}{d^2}, \quad \mu = \frac{\tau^2(\tau + 2)}{d(3 + \tau - \tau^2)}. \quad (77)$$

For $0 < \tau < 1$, all required positivity conditions hold. Both g and μ extend continuously to $[0, 1]$, taking values 0 and $1/6$ at the endpoints. Thus every gap and every central parameter in $(0, 1/6)$ is attained. Moreover $\tau^2 \leq 6g$ and $\tau^2 \leq 12\mu$ on this interval, so τ tends to zero as g or μ tends to zero. Consequently the limits obtained from (77) describe the entire central path tail, rather than a selected sequence of central points.

As $g \downarrow 0$, the formulas give

$$b \sim \sqrt{g/3}, \quad q \sim 2g/3, \quad \mu \sim 2g/3; \quad a \rightarrow 1, \quad b \sim \sqrt{\mu/2}, \quad q \sim \mu. \quad (78)$$

We now compute the Hessian in the four tangent coordinates (b, g, t, e) . The (b, g) and (t, e) subspaces are orthogonal in the Frobenius metric, and the Hessian has no mixed terms between them at $t = e = 0$, either by the same sign symmetry or by direct differentiation.

Let $A_2 = \begin{pmatrix} a & b \\ b & g \end{pmatrix}$, with determinant q . For an off-block tangent perturbation $U = \begin{pmatrix} 0_{2 \times 2} & v \\ v^\top & 0 \end{pmatrix}$, where $v = (t, e)^\top$, one has

$$D^2F(X)[U, U] = \text{tr}(X^{-1}UX^{-1}U) = \frac{2}{b}v^\top A_2^{-1}v, \quad \|U\|_F^2 = 2v^\top v.$$

Thus this block's eigenvalues are those of A_2^{-1}/b . The larger eigenvalue of A_2 tends to one, and the smaller is asymptotic to $q \sim \mu$, because their product is q . This gives the eigenvalues $\sqrt{2}\mu^{-1/2}$ and $\sqrt{2}\mu^{-3/2}$ to leading order.

On the (b, g) subspace, the tangent Gram matrix and the coordinate Hessian K are

$$G = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \quad K = \begin{pmatrix} b^{-2} + q_b^2/q^2 + 2/q & q_b q_g/q^2 + 1/q \\ q_b q_g/q^2 + 1/q & q_g^2/q^2 + 2/q \end{pmatrix}, \quad (79)$$

where $q_b = -g - 2b$ and $q_g = 1 - b - 2g$. The coordinates (b, g) are not orthonormal, so the eigenvalues of the operator in the Frobenius metric are not those of K ; they solve $Kv = \lambda Gv$. From (78),

$$\mu K_{bb} \rightarrow 6, \quad \mu^{3/2} K_{bg} \rightarrow -\sqrt{2}, \quad \mu^2 K_{gg} \rightarrow 1. \quad (80)$$

It follows that $\mu^2 G^{-1/2} K G^{-1/2}$ converges to the rank-one matrix $G^{-1/2} e_2 e_2^\top G^{-1/2}$, whose nonzero eigenvalue is $(G^{-1})_{22} = 4/7$. Hence the larger generalized eigenvalue is asymptotic to $(4/7)\mu^{-2}$. Also

$$\mu^3 \det K \rightarrow 6 - 2 = 4, \quad \det G = 7.$$

The product of the generalized eigenvalues is $\det K / \det G$, so the smaller is asymptotic to μ^{-1} . The four distinct powers determine their eventual increasing order, proving (75). Their extreme ratio proves (76), using $g \sim 3\mu/2$. Restricting to P_0 retains precisely the (b, g) block and gives its two stated eigenvalues and ratio. $\square$

The log-barrier calculation in [theorem 7.2](#) has a Lean formalization in the accompanying repository. It proves that the matrix is the unique minimizer of the barrier objective over all positive-definite feasible matrices, computes the second Fréchet derivative of $-\log \det$, identifies its operator in the

Frobenius metric, and factors its characteristic polynomial. It also proves the ordered asymptotics and both condition-number constants, for the restricted problem as well, and for the whole central path at small gaps. The formalization covers only the log-barrier spectrum. The singularity-degree and diameter claims in [proposition 7.1](#), and the extension to other barriers below, are not formalized.

The equal-gap Loewner comparison in [theorem 3.2](#) determines the order of every eigenvalue, not only the condition number. In the same Frobenius metric, every fixed barrier on P has ordered eigenvalues of orders $g^{-1/2}$, g^{-1} , $g^{-3/2}$, and g^{-2} , and every fixed barrier on P_0 has eigenvalues of orders g^{-1} and g^{-2} . The leading constants in (75) are for the canonical log barrier; we do not claim that other barriers have the same constants.

Spectral clustering in SDP has been studied before. In particular, [Alizadeh et al. \(1998, Section 4\)](#) analyze eigenvalue behavior of a primal–dual Schur matrix, and [Peña \(2002, Section 3.3\)](#) compares the conditioning of central-path systems. Those results concern a different matrix, and the cited SDP clustering analysis assumes strict complementarity. [Wei and Wolkowicz \(2010\)](#) construct instances with prescribed failure of strict complementarity and investigate their numerical difficulty.

[Sremac et al. \(2021, Section 4\)](#) relate singularity degree to spectral rates along an external regularization path: $X(\alpha)$ maximizes $\log \det X$ subject to $A(X) = b + \alpha A(B)$, with fixed $B \succ 0$, and $Z(\alpha) = \alpha X(\alpha)^{-1}$. Under their assumptions, Corollary 4.3 separates eigenvalue rates of $X(\alpha)$ and $Z(\alpha)$, Theorem 4.4 characterizes degree one through linear decay of the vanishing primal eigenvalues, and Theorem 4.7 gives separated rates for dual diagonal blocks. These are earlier results linking singularity degree to spectra. In contrast, our affine constraints are fixed, our path parameter is the primal objective gap, and our operator is the Hessian reduced to the equality tangent space. The eigenvalues of the primal iterate do not by themselves determine this reduced Hessian.

To our knowledge, the four leading constants in (75) have not been stated before for this trace-normalized classical construction.

8 LP spectral scales and the Newton right-hand side

The diameter law determines the ratio of the extreme eigenvalues. For a polytope, the dimension of the optimal face also determines how many eigenvalues have each of two orders. For the logarithmic barrier this splitting is classical active-set geometry, closely related to the Hessian analyses of [Wright \(1994\)](#) and [Forsgren et al. \(2002\)](#). Using [theorem 3.2](#), we show that every fixed finite-parameter barrier has the same ordered eigenvalue orders. The eigenspaces need not coincide, and their different rates of convergence matter for the Newton equation.

8.1 The optimal-face tangent and the two spectral scales

We use the compact, strictly feasible standard-form representation (52), now allowing an optimal face Φ of any dimension f . Let $d = n - m$, and identify $\mathcal{V} = \ker A$ with $\mathbb{R}^d$ using the orthonormal matrix W . Set

$$B = \{i : x_i > 0 \text{ for some } x \in \Phi\}, \quad N = \{1, \dots, n\} \setminus B, \quad \mathcal{T} = \{z \in \ker A : z_N = 0\}.$$

Then $\mathcal{T}$ is exactly the tangent space of $\text{aff } \Phi$, and $\dim \mathcal{T} = f$. Indeed, a relative-interior point x^a of Φ is positive on B . Hence, for every sufficiently short z supported on B with $Az = 0$, both $x^a \pm z$ are feasible. Optimality then forces the objective increment to vanish, so both points lie in Φ . Conversely, every chord of Φ vanishes on N . The objective is orthogonal to this tangent: $c_V \perp \mathcal{T}$. Since the objective is nonconstant, $f < d$. Write $P_{\mathcal{T}}$ for the orthogonal projector onto $\mathcal{T}$.

Classical endpoint theory for the logarithmic central path (Adler and Monteiro, 1991; Güler, 1994; Halická, 1999) gives

$$x_B(\mu) \longrightarrow x_B^a > 0, \quad s_N(\mu) \longrightarrow s_N^a > 0, \quad (81)$$

where the limiting pair is strictly complementary, even when the LP is degenerate. Positivity is componentwise. The reduced log-barrier Hessian has the exact decomposition

$$\begin{aligned} H_{\log}(x(\mu)) &= H_N(\mu) + H_B(\mu), \\ H_N &= \mu^{-2} W^\top R_N^\top \text{Diag}(s_N^2) R_N W, \\ H_B &= W^\top R_B^\top \text{Diag}(x_B^{-2}) R_B W, \end{aligned} \quad (82)$$

with coordinate restrictions R_B, R_N . Thus $\ker H_N = \mathcal{T}$ under the stated identification, while $\|H_B\| = O(1)$.

Theorem 8.1 (Two scales for every fixed barrier). *Let F be any fixed ν -self-concordant barrier on P° . In increasing eigenvalue order, as $g \downarrow 0$,*

$$\lambda_j(\widehat{H}_F(g)) = \begin{cases} \Theta(1), & 1 \leq j \leq f, \\ \Theta(g^{-2}), & f < j \leq d. \end{cases} \quad (83)$$

The first case is empty when the optimum is unique. All constants may depend on the fixed instance and ν . For sufficiently small g , let $\Pi_F(g)$ be the orthogonal projector onto the eigenspace of the lowest f eigenvalues (the weak eigenspace); put $\Pi_F = 0$ if $f = 0$. Then

$$\|\Pi_F(g) - P_{\mathcal{T}}\| = O(g), \quad (84)$$

$$\|\Pi_{\log}(g) - P_{\mathcal{T}}\| = O(g^2). \quad (85)$$

In particular, the weak eigenspace converges to $\mathcal{T}$, although it need not equal $\mathcal{T}$ at positive gap.

Proof. The nonzero singular values of the fixed restriction $R_N W$ are bounded away from zero. By (81), the restriction of H_N to $\mathcal{T}^\perp$ has all its eigenvalues between $a\mu^{-2}$ and $b\mu^{-2}$, for fixed $a, b > 0$ and all sufficiently small μ . Also $0 \preceq H_B \preceq b'I$. Weyl's inequalities give $d - f$ eigenvalues of order μ^{-2} . The minimax principle applied to $\mathcal{T}$ bounds the lowest f eigenvalues by b' . Dikin containment in the compact set P gives the common lower bound $H_{\log} \succeq (\text{diam } P)^{-2}I$. Therefore those f eigenvalues are of order one. By (15), $g = \Theta(\mu)$.

At equal objective gaps, (23) gives constants $a_F, b_F > 0$ such that

$$a_F \widehat{H}_{\log}(g) \preceq \widehat{H}_F(g) \preceq b_F \widehat{H}_{\log}(g). \quad (86)$$

The eigenvalue minimax principle proves (83). Moreover, for another constant $a' > 0$,

$$\widehat{H}_F(g) \succeq a' g^{-2} P_{\mathcal{T}^\perp}. \quad (87)$$

If Q_F is an orthonormal basis for the weak eigenspace, its compressed Hessian has norm $O(1)$. Evaluating (87) on $Q_F u$ gives $\|P_{\mathcal{T}^\perp} Q_F u\| \leq O(g) \|u\|$.

For the logarithmic barrier the eigenvector equation gives a sharper estimate. Write $H_{\log} Q_{\log} = Q_{\log} \Lambda_w$, with $\|\Lambda_w\| = O(1)$. Then

$$H_N Q_{\log} = Q_{\log} \Lambda_w - H_B Q_{\log}, \quad \|H_N Q_{\log}\| = O(1).$$

The inverse of H_N on $\mathcal{T}^\perp$ has norm $O(g^2)$, so $\|P_{\mathcal{T}^\perp}Q_{\log}\| = O(g^2)$. For subspaces of the same dimension, the norm of the difference of their orthogonal projectors equals the largest sine of their principal angles, here $\|P_{\mathcal{T}^\perp}Q_F\|$. This identity follows by taking a singular value decomposition of the matrix of inner products of their orthonormal bases. It proves both asserted projector rates. This is the elementary spectral-subspace perturbation argument underlying the familiar Davis–Kahan estimates (Davis and Kahan, 1970). $\square$

The proof uses endpoint convergence only for the reference logarithmic barrier, not for the general barrier. By [theorem 3.2](#), the constants are uniform over barriers with uniformly bounded ν . The ordered eigenvalue scales and the general $O(g)$ projector estimate also hold at points whose Newton decrement (17) is at most a fixed $\rho < 1$ for some $\mu > 0$, with g the point’s own objective gap. The right-hand-side statement below also requires exact centrality.

8.2 The exact Newton right-hand side and discarding the weak eigenspace

At an exact center $x_F(\mu_0)$, the Newton equation for $F(x) + \langle c, x \rangle / \mu_1$, with new parameter μ_1 , is

$$H_F(x_F(\mu_0)) \Delta x = -(\mu_1^{-1} - \mu_0^{-1})c_V. \quad (88)$$

The path derivative has a right-hand side in the same direction, since $H_F(x_F(\mu))x'_F(\mu) = \mu^{-2}c_V$. Thus, for every nonzero right-hand side r in (88),

$$\frac{\|\Pi_F(g)r\|}{\|r\|} = O(g), \quad \frac{\|\Pi_{\log}(g)r\|}{\|r\|} = O(g^2). \quad (89)$$

These follow immediately from $P_{\mathcal{T}}c_V = 0$ and [theorem 8.1](#). The quantities in (89) are ratios of norms; the corresponding ratios of squared norms are $O(g^2)$ and $O(g^4)$. At an approximately central point, the right-hand side has an additional centering residual, and spectral clustering alone does not give these estimates for it.

For positive-definite H and an orthogonal spectral projector Π of H , discarding the range of Π gives

$$u_s = H^{-1}(I - \Pi)r, \quad r - Hu_s = \Pi r, \quad H^{-1}r - u_s = H^{-1}\Pi r. \quad (90)$$

The relative Euclidean residual is thus exactly $\|\Pi r\| / \|r\|$, but the relative solution error is not bounded by this fraction. Now suppose, in the LP setting, that the spectral projection and a representation of H restricted to the complementary eigenspace $\text{range}(I - \Pi)$ of the large eigenvalues are available. By [theorem 8.1](#), this restricted operator has a bounded condition number. If a computed restricted solution satisfies

$$\tilde{u}_s \in \text{range}(I - \Pi), \quad \|(I - \Pi)r - H\tilde{u}_s\| \leq \delta \|(I - \Pi)r\|,$$

then orthogonality gives

$$\frac{\|r - H\tilde{u}_s\|^2}{\|r\|^2} \leq \frac{\|\Pi r\|^2}{\|r\|^2} + \delta^2. \quad (91)$$

This bound ignores the cost of constructing the projection and does not show that a path-following algorithm’s own error criterion is met. A Euclidean residual bound alone implies neither accurate Newton directions nor convergence of an inexact interior-point method.

8.3 A sharp general-barrier example

The map $(x, y) \mapsto (1 + x, 1 - x, y, 1 - y)$ embeds the rectangle below in a strictly feasible standard form. It multiplies the tangent metric by two, so it preserves eigenvector angles and all gap exponents. The example therefore tests the rates of [theorem 8.1](#) directly.

Oscillating perturbations of self-concordant barriers are not new: [Duistermaat \(2001, Remark 2.3\)](#) uses $f + A \sin f$ to produce rescaled derivatives without boundary limits. We use the perturbation $x \sin(\log y)$ below, which couples the two coordinates, to show that the eigenspace rate is sharp and to compare the residual with the solution error.

The following finite-parameter barrier shows that the $O(g)$ norm rate cannot in general be improved to $O(g^2)$. It also shows that the small residual in [\(90\)](#) can come with a large solution error.

Proposition 8.2 (Sharp eigenspace rate, and a small residual with a large solution error). *On $P = [-1, 1] \times [0, 1]$, minimize y and set*

$$\begin{aligned} F(x, y) &= 4(F_0(x, y) + \varepsilon \psi(x, y)), \\ F_0(x, y) &= -\log(1 - x^2) - \log y - \log(1 - y), \\ \psi(x, y) &= x \sin(\log y), \quad \varepsilon = 1/100. \end{aligned} \tag{92}$$

This is a 20-self-concordant barrier. At gaps $g_k = \exp(-2\pi k)$, $k = 1, 2, \dots$, its exact center is $(0, g_k)$. For its weak spectral projector and $r = (0, 1)^\top$,

$$\|\Pi_F(g_k)r\| \sim \varepsilon g_k. \tag{93}$$

With $H = \hat{H}_F(g_k)$ and $u_s = H^{-1}(I - \Pi_F(g_k))r$,

$$\frac{\|r - Hu_s\|}{\|r\|} \sim \varepsilon g_k, \quad \frac{\|H^{-1}r - u_s\|}{\|H^{-1}r\|} \longrightarrow 1. \tag{94}$$

Proof. We first check the barrier assumptions on the entire open rectangle. The four affine logarithms comprising F_0 make it a standard self-concordant barrier with parameter at most 4. Its Hessian is

$$H_0 = \text{Diag}(A_0, B_0), \quad A_0 = \frac{2(1 + x^2)}{(1 - x^2)^2} \geq 2, \quad B_0 = y^{-2} + (1 - y)^{-2} \geq y^{-2}.$$

For $h = (u, v)$ put $U = \sqrt{A_0}u$, $V = \sqrt{B_0}v$, so that $\|h\|_{H_0}^2 = U^2 + V^2$, $|u| \leq |U|/\sqrt{2}$, and $|v|/y \leq |V|$. Writing $s = \sin(\log y)$ and $t = \cos(\log y)$, direct differentiation gives

$$\begin{aligned} D\psi[h] &= su + xtv/y, \\ D^2\psi[h, h] &= 2tuv/y - x(s + t)v^2/y^2, \\ D^3\psi[h, h, h] &= -3(s + t)uv^2/y^2 + x(3s + t)v^3/y^3. \end{aligned}$$

Since $|x| < 1$, $|s + t| \leq \sqrt{2}$, and $|3s + t| \leq \sqrt{10}$, these imply

$$\begin{aligned} |D\psi[h]| &\leq 2\|h\|_{H_0}, \\ |D^2\psi[h, h]| &\leq 3\|h\|_{H_0}^2, \\ |D^3\psi[h, h, h]| &\leq 7\|h\|_{H_0}^3. \end{aligned} \tag{95}$$

For example, the second bound follows from $\sqrt{2}|UV| + \sqrt{2}V^2 \leq 3(U^2 + V^2)$; the third follows from $3|U|V^2 + \sqrt{10}|V|^3 \leq 7(U^2 + V^2)^{3/2}$. For $f_0 = F_0 + \varepsilon\psi$ we therefore have

$$D^2f_0[h, h] \geq (1 - 3\varepsilon)\|h\|_{H_0}^2, \quad |D^3f_0[h, h, h]| \leq (2 + 7\varepsilon)\|h\|_{H_0}^3, \quad |Df_0[h]| \leq (2 + 2\varepsilon)\|h\|_{H_0}.$$

The inequality $2 + 7\varepsilon \leq 4(1 - 3\varepsilon)^{3/2}$ proves standard self-concordance of $F = 4f_0$. Its barrier parameter is at most

$$\frac{4(2 + 2\varepsilon)^2}{1 - 3\varepsilon} < 20. \quad (96)$$

Its Hessian is positive definite. The bounded perturbation ψ does not affect divergence at the boundary, completing the global check.

At $y = g_k$, $s = 0$ and $t = 1$. At $x = 0$ the first derivative in x vanishes, while

$$F_y(0, g_k) = 4 \left(-\frac{1}{g_k} + \frac{1}{1 - g_k} \right) < 0.$$

Consequently $\mu_k = -1/F_y(0, g_k) > 0$ makes the point stationary for $F + y/\mu_k$. Strict convexity makes it the unique exact center. The objective gap is g_k . Define $J = H/4$ to keep the scaling explicit:

$$J = \begin{pmatrix} 2 & \varepsilon/g_k \\ \varepsilon/g_k & g_k^{-2} + (1 - g_k)^{-2} \end{pmatrix}. \quad (97)$$

For $g = g_k \downarrow 0$, its eigenvalues satisfy $\lambda_+(J) \sim g^{-2}$ and $\lambda_-(J) \rightarrow 2 - \varepsilon^2$, as follows from its trace and determinant. A unit weak eigenvector v_- with positive first coordinate obeys

$$\frac{(v_-)_2}{(v_-)_1} = -\frac{2 - \lambda_-(J)}{\varepsilon/g} \sim -\varepsilon g.$$

It follows that $|v_-^\top r| \sim \varepsilon g$, whereas $|v_+^\top r| \rightarrow 1$. Restoring $H = 4J$ gives

$$\|H^{-1}\Pi_F r\| \sim \frac{\varepsilon g}{4(2 - \varepsilon^2)}, \quad \|H^{-1}(I - \Pi_F)r\| \sim \frac{g^2}{4}. \quad (98)$$

The weak component thus dominates the Euclidean solution norm, while the residual of u_s is exactly $\Pi_F r$. This proves (94). $\square$

The example also shows why [theorem 8.1](#) does not assume that a general central path converges. At every sufficiently small y , the unique solution of

$$\frac{2x}{1 - x^2} + \varepsilon \sin(\log y) = 0$$

has $F_y(x, y) < 0$ and hence lies on the exact path for a positive μ . Indeed, $F_y/4 \leq (-1 + \varepsilon)/y + 1/(1 - y) < 0$ for small y . The solution x takes distinct limiting values along sequences with $\sin(\log y) = 1$ and -1 . Thus the path has no single endpoint, although the barrier parameter is finite. The weak eigenspace still converges to $\mathcal{T}$, as the theorem asserts.

Nonconvergent central paths were already constructed by [Bolte and Pauwels \(2022\)](#) using a Legendre function that is finite and continuous on a closed square. That function does not tend to infinity at the boundary, as [definition 2.1](#) requires. Our example gives nonconvergence for a self-concordant barrier whose parameter is bounded on the whole rectangle by (96), together with the sharp projector and solution-error estimates.

9 What spectral clustering implies for linear solves

When the condition number κ is large, the usual bound of $O(\sqrt{\kappa})$ conjugate-gradient (CG) iterations need not be sharp. Classical estimates for clustered spectra exploit the spectrum more precisely; see [Axelsson \(1994\)](#). We give the elementary polynomial construction that applies to the two-scale spectra of [theorem 8.1](#). It bounds the energy-norm error in exact arithmetic, not residuals in finite precision.

Proposition 9.1 (An explicit two-interval CG bound). *Suppose $H \succ 0$ and*

$$\text{spec}(H) \subseteq [a_1, b_1] \cup [a_2, b_2], \quad 0 < a_1 \leq b_1 < a_2 \leq b_2.$$

Put $\rho_i = b_i/a_i$ and $\Lambda = b_2/a_1$. For $0 < \delta \leq 1/2$ there is a real polynomial q , with $q(0) = 1$, such that $\max_{\text{spec}(H)} |q| \leq \delta$ and

$$\deg q = O(\sqrt{\rho_1 \rho_2} \log(4\Lambda) \log(1/\delta) + \sqrt{\rho_2} \log(1/\delta)). \quad (99)$$

Exact-arithmetic CG for $Hu = r$, from any initial vector u_0 , therefore achieves $\|u - u_k\|_H \leq \delta \|u - u_0\|_H$ within this many iterations, or within $\dim H$ iterations if that is smaller.

Proof. For $0 < a < b$ the normalized Chebyshev polynomial

$$r_m(x; a, b) = \frac{T_m((a + b - 2x)/(b - a))}{T_m((a + b)/(b - a))}$$

has value one at zero and satisfies

$$\max_{[a, b]} |r_m| \leq 2 \left(\frac{\sqrt{b/a} - 1}{\sqrt{b/a} + 1} \right)^m \leq 2 \exp \left(-\frac{2m}{\sqrt{b/a}} \right). \quad (100)$$

The first bound follows from $|T_m| \leq 1$ on $[-1, 1]$ and $T_m(t) = (w^m + w^{-m})/2$ for $t = (w + w^{-1})/2 > 1$. The second uses $\log((s + 1)/(s - 1)) \geq 2/s$ for $s > 1$.

Choose $p_2 = r_{m_2}(\cdot; a_2, b_2)$ with $m_2 = \lceil \sqrt{\rho_2} \rceil$. Its magnitude is at most $1/2$ on the upper interval, and at most one on $[0, a_2]$, because $T_{m_2}(t)$ is positive and increasing for $t \geq 1$. If $a_2 = b_2$, use $p_2(x) = 1 - x/a_2$, $m_2 = 1$ instead. Choose $r_1 = r_{m_1}(\cdot; a_1, b_1)$ with $m_1 = \lceil \frac{1}{2} \sqrt{\rho_1} \log(2/\delta) \rceil$, so that $|r_1| \leq \delta$ on the lower interval. If $a_1 = b_1$, use $r_1(x) = 1 - x/a_1$ and $m_1 = 1$.

All m_1 roots θ_j of r_1 lie in $[a_1, b_1]$, and $r_1(0) = 1$, so its product representation gives

$$|r_1(x)| = \prod_{j=1}^{m_1} |1 - x/\theta_j| \leq \Lambda^{m_1} \quad (a_2 \leq x \leq b_2).$$

Let

$$j = \lceil m_1 \log_2(4\Lambda) + \log_2(1/\delta) \rceil, \quad q = r_1 p_2^j.$$

On the lower interval $|q| \leq \delta$; on the upper interval $|q| \leq \Lambda^{m_1} 2^{-j} \leq \delta$. Its degree $m_1 + jm_2$ gives (99). Finally, exact-arithmetic CG minimizes the energy-norm error over vectors $u - u_k = q_k(H)(u - u_0)$ with $q_k(0) = 1$ and degree at most k . The spectral theorem bounds the relative energy error of our candidate by $\max_{\text{spec}(H)} |q|$. The usual exact termination in dimension $\dim H$ gives the stated alternative. $\square$

For each fixed barrier in [theorem 8.1](#), the two interval ratios are bounded and $\Lambda = \Theta(g^{-2})$ when $f > 0$. Thus [proposition 9.1](#) gives $O(\log(1/g) \log(1/\delta))$ iterations for all sufficiently small g . If $f = 0$, there is one interval with bounded condition number, and the ordinary Chebyshev estimate gives $O(\log(1/\delta))$. These bounds count matrix–vector products, not the cost of computing them. In floating point, loss of conjugacy can change both the attainable error and the iteration count, so a numerical use of the bound should check a recursively updated residual against a freshly computed $r - Hu_k$.

9.1 Quantum access and output are separate assumptions

A quantum linear-system algorithm usually receives a *block encoding* of H : a unitary whose specified block equals H/α , up to a controlled encoding error, with normalization $\alpha \geq \|H\|$. The effective condition number of the encoding is

$$\kappa_{\text{enc}} = \alpha \|H^{-1}\|, \quad (101)$$

which can exceed $\kappa(H)$. The algorithm also needs a procedure that prepares the normalized right-hand side, and a specified output, often the normalized solution state rather than all its entries. These assumptions are part of the problem, not consequences of Hessian geometry.

Quantum singular value transformation applies suitably bounded polynomials to a normalized encoded matrix ([Gilyén et al., 2019](#)). The boundedness condition must hold on the whole interval $[-1, 1]$, including points outside the spectrum. In contrast, the polynomial in [proposition 9.1](#) is controlled only on the two spectral intervals, and its values in the gap between them are unrestricted. This construction alone therefore does not give a quantum algorithm whose query cost equals the CG degree.

This is consistent with the lower bounds and structured positive results of [Orsucci and Dunjko \(2021\)](#). Their Proposition 6 proves a worst-case $\Omega(\min\{\kappa, N\})$ query lower bound for positive-definite quantum linear systems of dimension N , with their matrix and state-preparation oracles. This bound is linear in κ only when N is at least of order κ . Such a worst-case family is not by itself an accuracy lower bound along one central path of fixed dimension. Their faster positive-definite algorithms require additional normalization, decomposition, or state-overlap conditions. Merely writing H as a Gram matrix does not give a square-root dependence on κ in general.

The right-hand side also matters. Our geometric theorems concern all eigenvalues, whereas exact path-following uses the particular right-hand side in (88). The residual bound (91) exploits it only under its stated projection and solve assumptions. Moreover, [proposition 8.2](#) shows that discarding a small fraction of the right-hand side can lose almost all of the Euclidean solution norm. A residual guarantee therefore cannot simply replace a guarantee on the quantum solution state.

Finally, Hessian conditioning does not obstruct every quantum approach to optimization. [Apers and Gribling \(2026\)](#) give quantum speedups for tall LPs with row access, using spectral approximations and gradient estimation to return an explicit feasible approximate solution. Their method avoids the Hessian-condition-number dependence of directly substituting a quantum linear-system solver. Our results describe a fixed metric and its spectrum; they neither rule out that algorithm nor prove an end-to-end separation between classical and quantum running times.

10 Metrics, formulations, and the need for a finite barrier parameter

10.1 Fixed coordinate changes and gap-dependent preconditioning

All spectral rates so far use the fixed Euclidean metric on the feasible affine hull. If $R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a fixed invertible change of tangent coordinates, the Hessian matrix becomes $R^\top H R$. With Euclidean

matrix condition numbers in each coordinate system,

$$\frac{\kappa(H)}{\kappa(R)^2} \leq \kappa(R^\top H R) \leq \kappa(R)^2 \kappa(H). \quad (102)$$

The upper bound follows by bounding the extreme Rayleigh quotients using the extreme singular values of R ; the lower bound follows by applying the upper bound to R^{-1} . Thus fixed coordinate changes preserve the exponents of g , possibly with large constant factors. If instead the Euclidean metric is transported, its Gram matrix is $R^\top R$, and the generalized spectrum of $(R^\top H R, R^\top R)$ is exactly the spectrum of H . A gap-dependent transformation is different: the idealized choice $R(g) = \widehat{H}_F(g)^{-1/2}$ makes the transformed matrix the identity. Whether such a preconditioner is useful depends on the cost of constructing, applying, and representing it. Neither the geometric theorems nor (102) bound these costs.

10.2 The reduced Hessian is not the Schur or augmented matrix

Let $B = \nabla^2 F(x) \succ 0$ be the *ambient* barrier Hessian for equality constraints $Ax = b$. The reduced matrix is $W^\top B W$ for an orthonormal basis W of $\ker A$; the primal Newton Schur matrix is $AB^{-1}A^\top$. The Schur matrix uses B , not the reduced matrix. Under an invertible change of ambient coordinates $z = Gx$, the barrier becomes $\tilde{F}(z) = F(G^{-1}z)$ and the constraint matrix becomes $\tilde{A} = AG^{-1}$. Then

$$\tilde{B} = G^{-\top} B G^{-1}, \quad \tilde{A} \tilde{B}^{-1} \tilde{A}^\top = AB^{-1}A^\top. \quad (103)$$

In contrast, the augmented Newton matrix transforms by congruence:

$$\begin{pmatrix} \tilde{B} & \tilde{A}^\top \\ \tilde{A} & 0 \end{pmatrix} = \begin{pmatrix} G^{-\top} & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} B & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} G^{-1} & 0 \\ 0 & I \end{pmatrix}. \quad (104)$$

Congruent matrices need not have the same spectrum, so a bound for the reduced Hessian is not the same bound for every Schur, scaled primal–dual, or augmented formulation. Classical conditioning analyses such as those of Peña (2002) and Renegar (1996) must be compared on the matrix they study.

For an explicit illustration of (103), consider the Lorentz cone and its barrier $F(t, z) = -\log(t^2 - \|z\|^2)$. For $0 < \lambda < 1$, let G_λ act on the first two coordinates as

$$\frac{1}{2} \begin{pmatrix} \lambda + \lambda^{-1} & \lambda - \lambda^{-1} \\ \lambda - \lambda^{-1} & \lambda + \lambda^{-1} \end{pmatrix}$$

and as the identity on the other coordinates. It preserves $t^2 - \|z\|^2$ and the positive sheet, so $F(G_\lambda x) = F(x)$. At $e_0 = (1, 0, \dots, 0)$ the ambient Hessian is $2I$, while at $G_\lambda e_0$ it is $2G_\lambda^{-2}$, with condition number λ^{-4} . Nevertheless, transforming A as above leaves the Schur matrix exactly unchanged. This only illustrates coordinates on the cone; our compact-set theorems are not applied to this unbounded cone. Varying λ can also change coefficient magnitudes and encoding normalizations, even though the transformation mixes only two coordinates.

The coordinate change $z = Gx$, with $\tilde{B}$ as in (103), also preserves the intrinsic length $\int \sqrt{\dot{x}^\top B(x) \dot{x}} dt$ of a differentiable curve. Its geometric role is studied by Nesterov and Todd (2002). This integral and a pointwise Euclidean condition number describe different quantities. Sensitivity of the center to the objective also requires an absolute scale. At fixed μ , implicit differentiation of exact centrality gives

$$D_{c_V} x_F(\mu) = -\mu^{-1} H_F(x_F(\mu))^{-1}. \quad (105)$$

Thus a lower Hessian eigenvalue bound controls this derivative; the condition number alone does not. A derivative with respect to a family parameter also contains the derivative of the data. For example, choose a short line segment $z_0 + [-a, a]v$ contained in a fixed barrier domain and set $z_k(t) = z_0 + a \sin(kt)v$. Continuity and positive definiteness give uniform spectral bounds for the Hessian on that segment, while $\|z'_k(0)\| = ak\|v\|$ is unbounded in k . Taking $c_k(t) = -\nabla F(z_k(t))$ makes $z_k(t)$ the exact center for $\mu = 1$ with objective $c_k(t)$; the data derivatives vary accordingly, consistently with (105). So pointwise spectral bounds do not control variation with respect to an arbitrary family parameter. The example does not show unbounded sensitivity per unit objective perturbation when the Hessian is uniformly bounded below. More generally, differentiating an auxiliary representation of a central point introduces derivatives of its data maps. Comparing this sensitivity with Newton conditioning requires bounds on those maps and on the Jacobian of the defining equations; pointwise centrality alone does not provide them.

10.3 Why a finite barrier parameter is essential

The containment lemma (lemma 3.1), and hence the diameter law, needs the gradient inequality (7). Differential self-concordance (6) and boundary divergence alone do not suffice. On the same rectangle as proposition 8.2, for fixed $\eta > 0$ consider

$$F_\eta(x, y) = -\log(1 - x^2) - \log y - \log(1 - y) + \frac{\eta}{y}. \quad (106)$$

This function is convex, diverges at the boundary, and is differentially self-concordant. To check the last assertion, for $h(y) = -\log y + \eta/y$ write $t = \eta/y \geq 0$. Then

$$h''(y) = y^{-2}(1 + 2t), \quad |h'''(y)| = 2y^{-3}(1 + 3t), \quad (1 + 3t)^2 \leq (1 + 2t)^3,$$

where the difference of the right and left sides is $3t^2 + 8t^3$. Adding the other affine logarithms preserves differential self-concordance, by summing their third-derivative bounds and using $\sum a_i^{3/2} \leq (\sum a_i)^{3/2}$ for $a_i \geq 0$.

For the objective y , the central path near the optimum has $x = 0$ and $g = y$. Indeed, $\partial_x F_\eta(0, g) = 0$ and $\partial_y F_\eta(0, g) = -1/g + 1/(1 - g) - \eta/g^2 < 0$ for small g , so $\mu(g) = -1/\partial_y F_\eta(0, g) > 0$ makes $(0, g)$ the unique center by strict convexity. The diagonal Hessian has entries 2 and $g^{-2} + (1 - g)^{-2} + 2\eta g^{-3}$, so

$$\widehat{\kappa}_{F_\eta}(g) \sim \eta g^{-3}, \quad \frac{|\partial_y F_\eta(0, g)|^2}{\partial_{yy} F_\eta(0, g)} \sim \frac{\eta}{2g} \longrightarrow \infty. \quad (107)$$

Hence no finite ν satisfies the barrier-gradient inequality. This does not contradict the $\Theta(g^{-2})$ law for finite-parameter barriers at a positive-dimensional optimal face. The same distinction between differential self-concordance and a finite global gradient parameter is noted explicitly for inverse-power perturbations in Monteiro and da Silva (2026, Remark 11.1); no result of that preprint is needed for the calculation here.

11 Reproducible numerical illustrations

The computations check explicit formulas and illustrate finite-gap behavior. They do not estimate an unproved asymptotic exponent or benchmark an end-to-end interior-point algorithm. The accompanying reproduction package contains all input arrays, exact rational certificates, scripts, raw rows including rejected points, generated figures and tables, and a manifest of commands, software versions, random seeds, transformations, and file hashes. Regeneration uses Python, NumPy, SciPy, and Matplotlib without network access or imports from another repository. High-precision scalar calculations and reference solves use the standard-library `decimal` module.

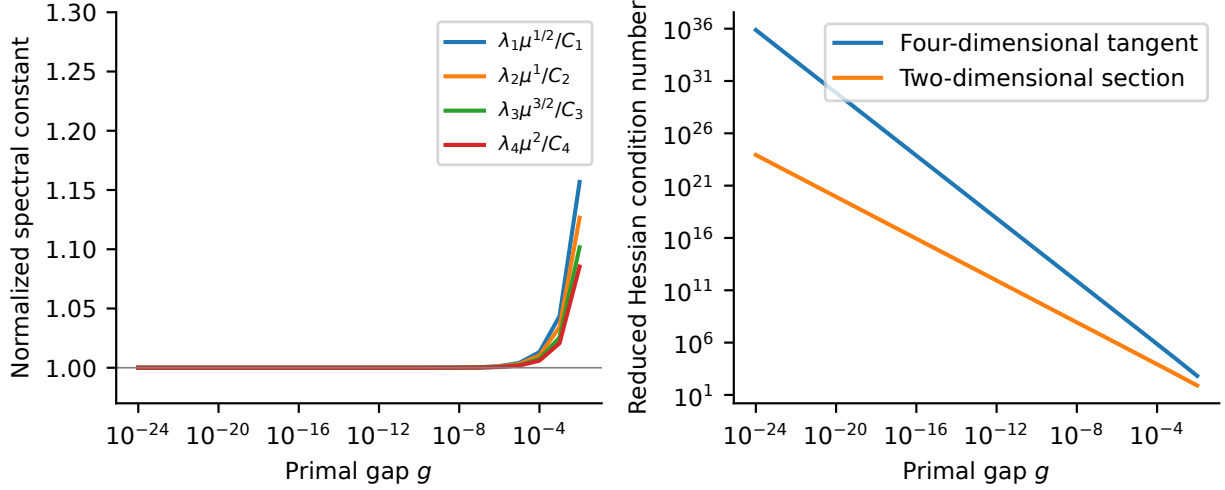


Figure 1: The fractional SDP and the restricted problem on P_0 . Left: the scaled ordered eigenvalues divided by $C = (\sqrt{2}, 1, \sqrt{2}, 4/7)$, with powers $(1/2, 1, 3/2, 2)$ of μ . Right: the condition numbers on the four-dimensional tangent space of the fractional SDP and on the two-dimensional tangent space of the restricted problem (legend: “Two-dimensional section”). Both optimality systems have singularity degree two.

11.1 Exact examples

Figure 1 evaluates the exact central-point formulas of [theorem 7.2](#), using a numerically stable quadratic-root formula, with 80 decimal digits, for $g = 10^{-2}, 10^{-3}, \dots, 10^{-24}$. The four scaled eigenvalues, divided by their proved limits, converge to one. The restricted problem on P_0 retains only the two (b, g) eigenvalues, so its condition-number exponent differs. At $g = 10^{-2}, 10^{-4}, 10^{-6}$, a separate float64 computation forms the Hessian through $\text{tr}(X^{-1}UX^{-1}V)$ and solves the generalized eigenproblem using the tangent Gram matrix. Its eigenvalues agree with the scalar formulas to relative error below 10^{-7} . The high-precision formulas avoid cancellation by computing the smaller root as the determinant divided by the larger root.

For the simplex of [proposition 5.4](#), the exact log-barrier center satisfies $x_i = \mu/(c_i + z)$, where $z > 0$ is the unique root of $\sum_i \mu/(c_i + z) = 1$. We solve this monotone scalar equation with 70 decimal digits and evaluate the Hessian through its factor. [Figure 2](#) shows the g^{-2} regime giving way to the plateau of order ϑ^{-2} when g falls below ϑ . For the compact degenerate LP of [example 6.2](#), direct centering gives, at $\mu = 10^{-9}$, $\mu^2\lambda_{\min} \approx 0.24888160$ and $\mu^2\lambda_{\max} \approx 1.14289068$, consistent with the proved limit and $\kappa(L) \approx 4.59210571$. The full sequence and its Newton decrements are supplied in the data package.

[Table 1](#) evaluates the exact centers at the gaps g_k in [proposition 8.2](#). It records the norm of the discarded right-hand-side component divided by g , and the relative Euclidean error of u_s , which discards the weak eigenspace. The discarded component tends to zero at the proved rate, while the solution error tends to one. This directly checks that a small residual need not give a small solution error; no iterative solver is involved.

11.2 Netlib instances: exact hypothesis checks and numerical acceptance tests

We use three small Netlib LPs ([Gay, 2013](#)): `afiro`, `sc50b`, and `adlittle`. The frozen arrays are the result of HiGHS 1.15.1 default presolve followed by conversion of inequalities and finite bounds to

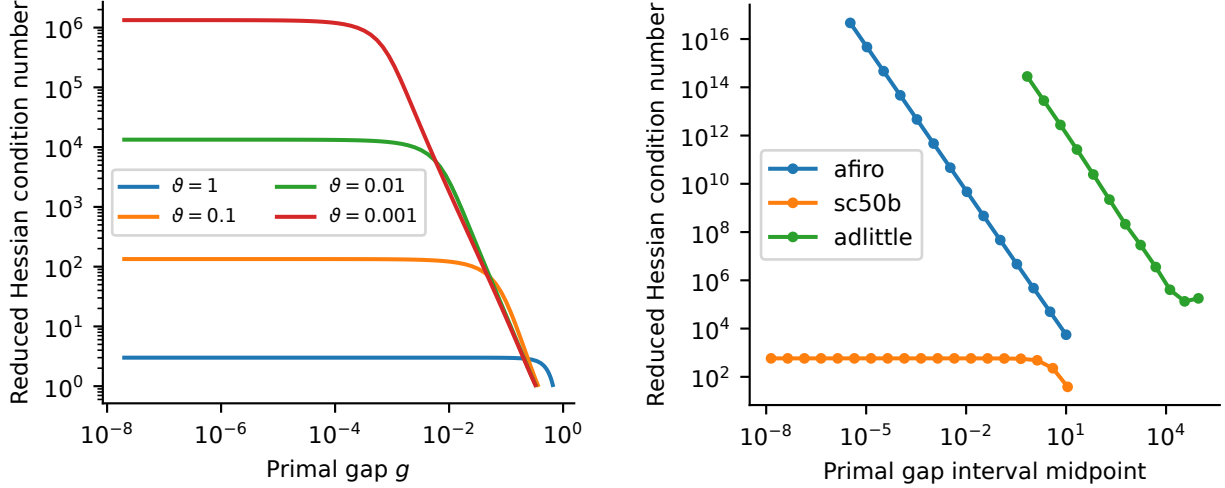


Figure 2: Left: near-degenerate simplex condition numbers for fixed values of ϑ . Right: accepted finite-gap points for the frozen Netlib representations. The right-hand curves are numerical observations; we fit no asymptotic exponent to them.

k	g_k	$\ \Pi c\ /(g_k \ c\)$	Relative solution error
1	1.87e-03	0.01000003	0.93680040
2	3.49e-06	0.01000000	0.99999976
3	6.51e-09	0.01000000	1.00000000
4	1.22e-11	0.01000000	1.00000000

Table 1: Oscillatory barrier of [proposition 8.2](#) with $\varepsilon = 0.01$, $g_k = \exp(-2\pi k)$, and $c = r = (0, 1)^\top$. With $\Pi = \Pi_F(g_k)$, the relative solution error is $\|H^{-1}\Pi c\|/\|H^{-1}c\|$. Entries shown as one are rounded.

equality constraints and nonnegative variables. Their dimensions, from original MPS to presolved model to standard form, are respectively $(27, 32) \rightarrow (7, 10) \rightarrow (9, 18)$, $(50, 48) \rightarrow (15, 15) \rightarrow (15, 29)$, and $(56, 97) \rightarrow (53, 95) \rightarrow (55, 136)$. The Hessian uses the Euclidean metric in these final coordinates. It is not the Hessian of the original MPS representation. The frozen arrays define the tested instances, so reproduction does not depend on the presolver version.

We check the hypotheses exactly, without feasibility tolerances. Each stored float64 input is read as its exact binary rational. The package verifies a rational point $x > 0$ with $Ax = b$ and a rational vector y with

$$A^\top y + \eta c > 0, \quad \eta \geq 0.$$

For feasible $x \geq 0$ in a fixed objective sublevel set, $x^\top (A^\top y + \eta c) = b^\top y + \eta c^\top x$ bounds every coordinate. The certificates use $\eta = 0$ for **afiro** and **sc50b**, proving compactness of their feasible sets, and $\eta = 1$ for **adlittle**, proving compactness of its objective sublevels. Thus the localized result, [proposition 3.5](#), applies to the last instance. Rational primal and dual feasible points also bracket the optimal value, with widths approximately $8.2 \cdot 10^{-12}$, $9.4 \cdot 10^{-12}$, and $4.5 \cdot 10^{-7}$, respectively.

Centering uses an orthonormal nullspace basis W and the Hessian factor $B = \text{Diag}(x^{-1})W$ (a matrix here, not the index set B of [section 6](#)). The Hessian eigenvalues are the squared singular values of B ; Newton solves and decrements use the same factorization. Each stored iterate is repaired to an exactly feasible rational point using a fixed nonsingular column basis. We evaluate its objective

Instance	(m, n)	Accepted/attempted	Last gap	κ	ρ
afiro	(9, 18)	14/27	3.34e-06	4.65e+16	2.2e-07
sc50b	(15, 29)	19/27	1.40e-08	5.82e+02	6.1e-07
adlittle	(55, 136)	12/27	6.80e-01	2.81e+14	2.5e-09

Table 2: Netlib results: accepted points out of 27 decreasing values of μ , and the gap, condition number κ of the reduced log-barrier Hessian, and Newton decrement $\rho = \rho_F(x; \mu)$ at the last accepted point. The raw data retain all attempted points, including rejected ones.

g	Iterations	Recursive residual	True residual	Relative energy error
1e-01	43	8.82e-09	8.82e-09	3.54e-09
1e-02	71	2.56e-09	2.56e-09	4.21e-09
1e-03	96	4.37e-09	4.39e-09	4.24e-09
1e-04	124	2.95e-09	2.52e-08	8.55e-09

Table 3: Float64 CG on the synthetic two-interval matrices. The final two columns use an 80-digit reference calculation for the rounded matrix and right-hand side. Passing the recursive stopping test does not guarantee a true relative residual below 10^{-8} .

exactly and subtract the optimal-value bracket. A point is accepted for the plot only when this gap interval is positive and has relative width at most 1%, the repaired point is positive, the maximum relative coordinate repair is at most 10^{-4} , the scaled equality residual is at most 10^{-12} , and the Newton decrement $\rho_F(x; \mu)$ of (17) is at most 10^{-5} . We additionally require

$$\epsilon_{\text{mach}} \max(n, d) \kappa_2(B) \leq 10^{-6}.$$

This last test indicates whether the SVD of the factor resolves its singular values numerically; it gives no rigorous error bounds. The exact rational certificates prove the instance hypotheses, but the plotted spectral values are floating-point computations.

Table 2 shows that the range of resolved gaps differs between instances. In particular, the spectral-resolution test stops **adlittle** well before its smallest attempted gap. We draw no conclusion about its limiting condition number from this truncation. The near-degenerate simplex already shows that a slope or plateau observed over a finite range does not by itself determine the asymptotic regime.

11.3 Clustered CG and stopping criteria

To test two intervals with many distinct eigenvalues, we generate a 60×60 symmetric matrix by a fixed-seed orthogonal rotation of 30 equally spaced values in $[1, 2]$ and 30 in $[g^{-2}, 3g^{-2}]$. This synthetic matrix is not a Newton Hessian. The float64 matrix is explicitly symmetrized. CG uses ordinary float64 matrix-vector products and stops when its recursively updated relative residual first reaches 10^{-8} . At that iterate, an 80-digit solve of the exact real system defined by the stored float64 matrix and right-hand side gives the reference solution, the relative energy error, and an independently evaluated true residual.

The prescribed endpoint ratios of the two intervals are two and three, while their separation grows as g decreases. These ratios describe the eigenvalues before rounding to float64, not the exact spectrum of the rounded matrix. Table 3 illustrates the benefit of the clustered structure and the difference between stopping criteria. It does not verify the exact-arithmetic degree bound empirically:

in exact arithmetic, CG terminates in at most 60 steps for these matrices, whereas in floating point it can take more steps because conjugacy is lost. We therefore report the recomputed residual and the energy error separately, as recommended in [section 9](#).

12 Discussion

At a point with Newton decrement below one, the Dikin ellipsoid of the barrier is determined, up to a factor depending only on the barrier parameter and the Newton decrement, by the difference body of the current objective sublevel set, which does not depend on the barrier. This gives the diameter law and the equal-gap comparison of all eigenvalues of different barriers. Within a fixed representation and metric, changing the finite-parameter barrier therefore cannot remove a conditioning exponent imposed by the sublevel sets; if the barrier varies with the requested gap, this needs a uniform bound on the barrier parameters.

The sublevel geometry determines the conditioning more precisely than uniqueness of the optimum, primal degeneracy, or singularity degree alone. Outside the polyhedral setting, a unique optimum does not by itself imply bounded conditioning. Conversely, in an LP, a degenerate primal optimal vertex does not prevent a positive-definite limit of the scaled log-barrier Hessian. In SDP, an attained fractional error bound can determine the conditioning exponent, but singularity degree alone cannot: the two problems of [section 7](#), whose optimality systems both have singularity degree two, have diameters $D(g)$ that shrink at different rates. The width bounds show why the ratio of the extreme eigenvalues need not describe the intermediate eigenvalues.

Solver cost also depends on the right-hand side and the required accuracy. Near the end of an LP central path, the exact Newton right-hand side has a small component in the eigenspace of the small eigenvalues, yet the oscillatory barrier of [proposition 8.2](#) shows that discarding this component can prevent accurate recovery of the Newton direction. In exact arithmetic, conjugate gradients can exploit a spectrum in separated clusters; the guarantee bounds the energy-norm error, which the numerical examples show must be distinguished from the recursively computed residual. Quantum applications need additional assumptions on data access and output before either fact gives a runtime bound.

These results concern one operator, the reduced Hessian in a fixed metric. Changes of representation, gap-dependent preconditioners, and primal–dual formulations can have different spectra and can be useful. Our comparison shows which conditioning the objective sublevel sets force before any linear algebra is accounted for.

References

- Ilan Adler and Renato D. C. Monteiro. Limiting behavior of the affine scaling continuous trajectories for linear programming problems. *Mathematical Programming*, 50(1–3):29–51, 1991. doi: 10.1007/BF01594923.
- Farid Alizadeh, Jean-Pierre A. Haeberly, and Michael L. Overton. Primal-dual interior-point methods for semidefinite programming: Convergence rates, stability and numerical results. *SIAM Journal on Optimization*, 8(3):746–768, 1998. doi: 10.1137/S1052623496304700.
- Simon Apers and Sander Gribling. Quantum speedups for linear programming via interior point methods. *SIAM Journal on Computing*, 55(1):93–134, 2026. doi: 10.1137/25M1736098.

- Owe Axelsson. *Iterative Solution Methods*. Cambridge University Press, 1994. doi: 10.1017/CBO9780511624100.
- Jérôme Bolte and Edouard Pauwels. Curiosities and counterexamples in smooth convex optimization. *Mathematical Programming*, 195:553–603, 2022. doi: 10.1007/s10107-021-01707-1.
- James V. Burke and Michael C. Ferris. Weak sharp minima in mathematical programming. *SIAM Journal on Control and Optimization*, 31(5):1340–1359, 1993. doi: 10.1137/0331063.
- Chandler Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970. doi: 10.1137/0707001.
- Dmitriy Drusvyatskiy and Henry Wolkowicz. The many faces of degeneracy in conic optimization. *Foundations and Trends in Optimization*, 3(2):77–170, 2017. doi: 10.1561/24000000011.
- Johannes J. Duistermaat. On the boundary behaviour of the riemannian structure of a self-concordant barrier function. *Asymptotic Analysis*, 27(1):9–46, 2001. doi: 10.3233/ASY-2001-448.
- Anders Forsgren, Philip E. Gill, and Margaret H. Wright. Interior methods for nonlinear optimization. *SIAM Review*, 44(4):525–597, 2002. doi: 10.1137/S0036144502414942.
- Robert M. Freund. On the primal-dual geometry of level sets in linear and conic optimization. *SIAM Journal on Optimization*, 13(4):1004–1013, 2003. doi: 10.1137/S1052623401393645.
- David M. Gay. Netlib linear programming test problems, 2013. URL <https://www.netlib.org/lp/data/>. Collection maintained by David M. Gay; accessed September 7, 2026.
- András Gilyén, Yuan Su, Guang Hao Low, and Nathan Wiebe. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pages 193–204, 2019. doi: 10.1145/3313276.3316366.
- Osman Güler. Limiting behavior of weighted central paths in linear programming. *Mathematical Programming*, 65(1–3):347–363, 1994. doi: 10.1007/BF01581702.
- Margaréta Halická. Analytical properties of the central path at boundary point in linear programming. *Mathematical Programming*, 84(2):335–355, 1999. doi: 10.1007/s101070050025.
- Bruno F. Lourenço. Amenable cones: Error bounds without constraint qualifications. *Mathematical Programming*, 186(1–2):1–48, 2021. doi: 10.1007/s10107-019-01439-3.
- Sergio Assuncao Monteiro and Fabricio Alves Barbosa da Silva. A perturbed q -Tsallis self-concordant barrier for spectrally robust semidefinite programming, 2026. URL <https://arxiv.org/abs/2606.04348v1>. arXiv:2606.04348v1, preprint.
- Yurii Nesterov. *Lectures on Convex Optimization*, volume 137 of *Springer Optimization and Its Applications*. Springer, 2 edition, 2018. doi: 10.1007/978-3-319-91578-4.
- Yurii Nesterov and Arkadii Nemirovskii. *Interior-Point Polynomial Algorithms in Convex Programming*. Society for Industrial and Applied Mathematics, 1994. doi: 10.1137/1.9781611970791.
- Yurii E. Nesterov and Michael J. Todd. On the riemannian geometry defined by self-concordant barriers and interior-point methods. *Foundations of Computational Mathematics*, 2:333–361, 2002. doi: 10.1007/s102080010032.

- Davide Orsucci and Vedran Dunjko. On solving classes of positive-definite quantum linear systems with quadratically improved runtime in the condition number. *Quantum*, 5:573, 2021. doi: 10.22331/q-2021-11-08-573.
- Javier Peña. Conditioning of convex programs from a primal-dual perspective. *Mathematics of Operations Research*, 26(2):206–220, 2001. doi: 10.1287/moor.26.2.206.10551.
- Javier Peña. Two properties of condition numbers for convex programs via implicitly defined barrier functions. *Mathematical Programming*, 93(1):55–75, 2002. doi: 10.1007/s101070200294.
- James Renegar. Condition numbers, the barrier method, and the conjugate-gradient method. *SIAM Journal on Optimization*, 6(4):879–912, 1996. doi: 10.1137/S105262349427532X.
- James Renegar. *A Mathematical View of Interior-Point Methods in Convex Optimization*. Society for Industrial and Applied Mathematics, 2001. doi: 10.1137/1.9780898718812.
- Stefan Sremac, Hugo J. Woerdeman, and Henry Wolkowicz. Error bounds and singularity degree in semidefinite programming. *SIAM Journal on Optimization*, 31(1):812–836, 2021. doi: 10.1137/19M1289327.
- Jos F. Sturm. Error bounds for linear matrix inequalities. *SIAM Journal on Optimization*, 10(4):1228–1248, 2000. doi: 10.1137/S1052623498338606.
- Hayato Waki and Masakazu Muramatsu. Facial reduction algorithms for conic optimization problems. *Journal of Optimization Theory and Applications*, 158(1):188–215, 2013. doi: 10.1007/s10957-012-0219-y.
- Hua Wei and Henry Wolkowicz. Generating and measuring instances of hard semidefinite programs. *Mathematical Programming*, 125(1):31–45, 2010. doi: 10.1007/s10107-008-0256-3.
- Margaret H. Wright. Some properties of the hessian of the logarithmic barrier function. *Mathematical Programming*, 67(1–3):265–295, 1994. doi: 10.1007/BF01582224.
- Margaret H. Wright. Ill-conditioning and computational error in interior methods for nonlinear programming. *SIAM Journal on Optimization*, 9(1):84–111, 1998. doi: 10.1137/S1052623497322279.
- Stephen J. Wright and Dominique Orban. Properties of the log-barrier function on degenerate nonlinear programs. *Mathematics of Operations Research*, 27(3):585–613, 2002. doi: 10.1287/moor.27.3.585.312.
- Zikai Xiong and Robert M. Freund. The role of level-set geometry on the performance of PDHG for conic linear optimization, 2024. URL <https://arxiv.org/abs/2406.01942>. Version of July 15, 2024.